



Smart Spot

Technical Guide

v1.18



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1. Introduction

1.1. Overview

The Smart Spot device developed by Libelium enables air quality and crowd monitoring solutions. Smart Spots are configurable IoT devices that can monitor various environmental factors, such as air quality (gases and suspended particles), temperature, humidity and noise, as well as integrating complete weather stations.

Combining all these sensors and capabilities in a single device provides savings in installation, maintenance, management and communications.



Figure: Smart Spot device

1.2. Device models

Smart Spot supports the following models for different main purposes:

- **Smart Spot Air:** For air quality index purposes (including gas sensors, PM1, PM2.5, PM4, PM10, TSP, TPC sensors and others), allowing the integration of complex extensions such as noise level sensors or weather stations, with multiple communication and integration protocols available.
- **Smart Spot Air LoRaWAN:** For air quality index purposes (including gas sensors, PM1, PM2.5, PM4, PM10, TSP, TPC sensors and others), allowing the inclusion of basic ambient sensor extensions with LoRaWAN as the main communication method.
- **Smart Spot Crowd:** Smart device detection using WiFi and BLE using non-invasive and respectful techniques with multiple communication and integration protocols available.

As these models can be extended with additional sensors, a list of available combinations for each model is provided in the 'Smart Spot extensions' section for specific compatibility and availability information.

1.3. Connectivity

The 'Air' and 'Crowd' models offer 4G, GSMGPRS and NBloT as the main communication methods, with LWM2M as the device management protocol and MQTT or SENTILO for third party integration.

The Air LoRaWAN model provides LoRaWAN as the main communication method with full reporting and partial device management available. It allows the use of 4G, GSMGPRS and NBloT as backup communication methods for full device management and alternative integration methods.

WiFi communication is always included and available for limited coverage or indoor scenarios. Furthermore, these devices are FIWARE Ready through MQTT and LoRaWAN protocols.

1.4. Additional Services

Although the Smart Spot device supports additional protocols for reporting data to third-party platforms such as MQTT, SENTILO or Cayenne LPP, its primary method of remote interaction is the Lightweight M2M (LWM2M) protocol. It uses this protocol to communicate with Libelium's cloud server, where various services for the device are hosted.

These services enable various management, maintenance, monitoring and data enhancement tasks to be performed on the device throughout its lifecycle, from initial setup and factory configuration after manufacture to remote monitoring of its operation during real deployment.

Refer to **Application Note 130: Monitoring, Maintenance, and Continuous Improvement Services** for extended information about the services used by the Smart Spot.

1.5. Internal storage

The Smart Spot device has the ability to store measurements retrieved from its sensors or services internally, to prevent data loss in the event of problems with device connectivity or integration with third party platforms.

The main features for the internal memory are:

- 6.5 MB of storage capacity (data is stored in binary format for maximum optimization)
- Available for MQTT, LoRaWAN and HTTP third party integration protocols
- Available for all models, 'Crowd' and 'Air'

For more information see '[Data Retrieval methods](#)' section.

1.6. Power supply

To be scalable in different types of scenarios and territories, Smart Spots allow different types of outdoor and indoor power sources. In addition, as an optional feature, a high-capacity battery is available to provide autonomy against power shutdowns (for instance in street lamps).

1.7. Device Management platform

Libelium [Homard](#) platform allows the user to access the device configuration.

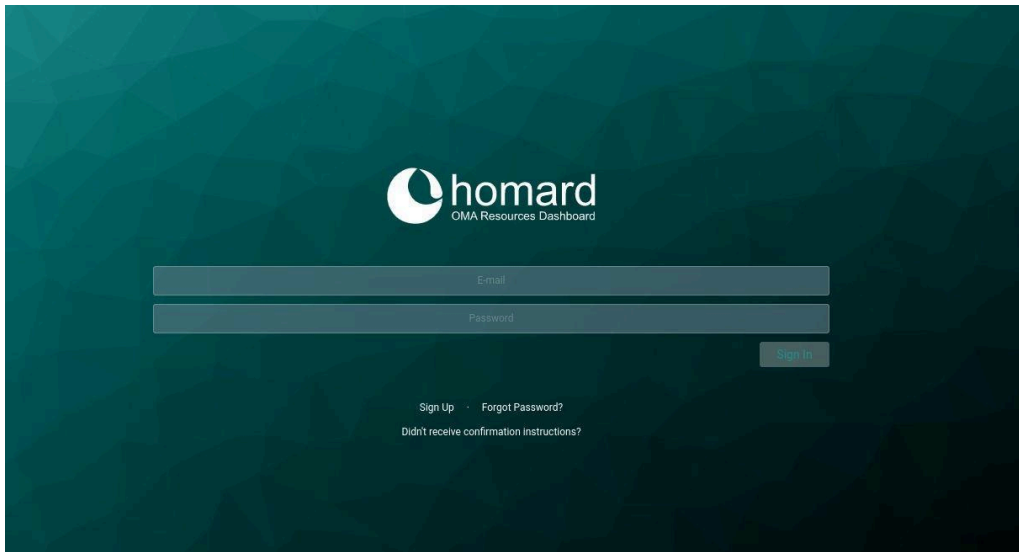


Figure: Libelium Homard login panel

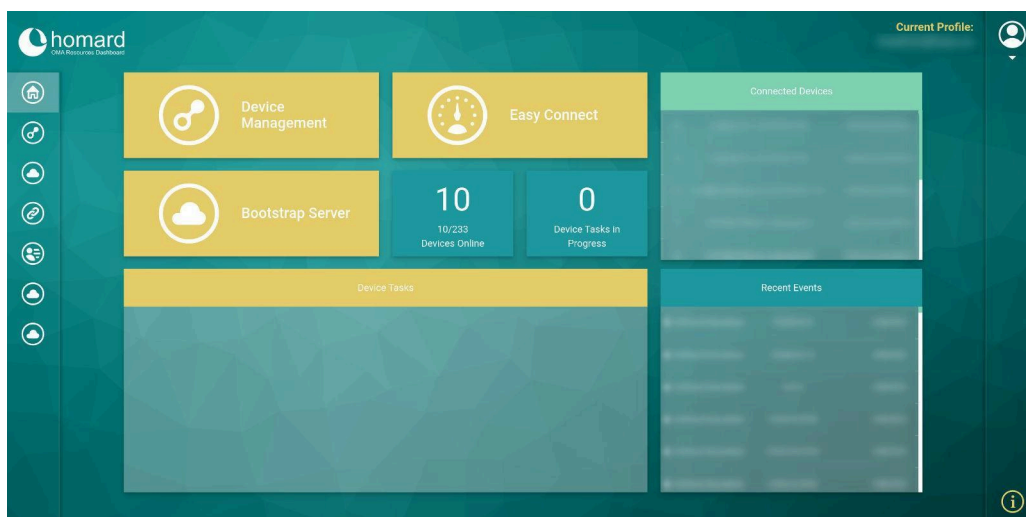


Figure: Libelium Homard platform

1.8. Visualizers

Libelium provides customized visualizers depending on the application deployed (Crowd monitoring, Air Quality pollution, Noise Level monitoring, etc.)

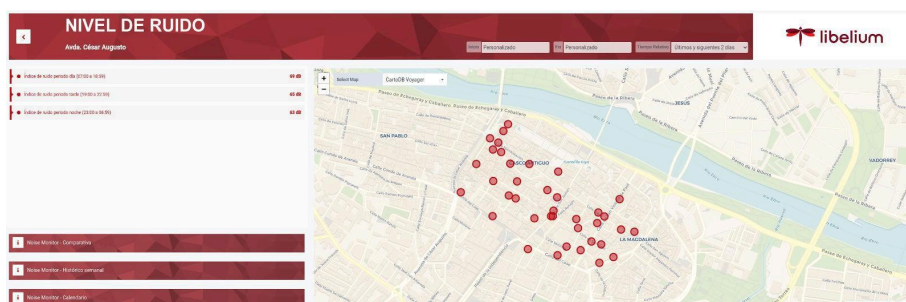


Figure: Custom visualizer

2. Smart Spot device

2.1. Technical specifications

2.1.1. Models

Model	Description
Air	Air Quality and weather sensors
Air LoRaWAN	Air Quality and basic ambient sensors with
LoRaWAN Crowd	WiFi and BLE scanning for device detection

2.1.2. Characteristics

Feature	Description
Operating system	Industrial operating system in real time (FreeRTOS)
Operating temperature range	-30°C to 60°C
CPU	Dual Core a 240 MHz. 16MB RAM / 32MB Flash
Internal memory	6.5 MB non-volatile memory for internal storage (available for 'Crowd' model only)
Antennas	Multi-antenna IP68 anti-vandalism (GPS/M2M/WiFi)
4G Antenna peak gain	4.87 dBi
WiFi Antenna peak gain	4.05 dBi
GPS Antenna peak gain	28±2 dBi (available only for '2G', '4G')
LoRaWAN Antenna peak gain	3 dBi (available for 'Air LoRaWAN' model only)
Antenna impedance	50 ohm nominal
Antenna operating frequencies	4G Main: 824-960/1710-2690 MHz 4G DiV: 1710-2690 MHz WiFi: 2400-2500 MHz GNSS: 1561/1575.42 MHz; LoRaWAN: 863-928MHz (available for 'Air LoRaWAN' model only)
Internal sensors	Temperature and Humidity
Vandalism detection	Accelerometer and Gyroscope

2.1.3. Communications

Feature	Description
Network	WiFi, 4G, 2G, NB-IoT (for both Remote management and Data sending)
Network ('Air LoRaWAN' model)	LoRaWAN
Protocols	MQTT Sentilo Cayenne LPP Extended (only available in 'Air LoRaWAN' model for data sending)
Remote management (full)	Full configuration management via Libelium
Remote management (partial)	Homard platform (LwM2M protocol) using WiFi, 4G, 2G, NB-IoT Partial configuration management via Third party platform (MQTT protocol or LoRaWAN (only available in 'Air LoRaWAN' model for data sending))

Data sending	Configurable between 1 min - 25 min
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2.1.4. LoRaWAN Specifications

Parameter	Description
LoRaWAN specification	1.0.2 LoRaWAN
Version Frequency band	EU868 (EU863-870)
LoRaWAN Class	Class A
Join method	OTAA
Adaptive Data Rate	Configurable (enabled by default)

Supported Data Rates:

Data Rate	Configuration
0	SF12 / 125 kHz
1	SF11 / 125 kHz
2	SF10 / 125 kHz
3	SF9 / 125 kHz
4	SF8 / 125 kHz
5	SF7 / 125 kHz
6	SF7 / 250 kHz
7	FSK: 50 kbps

⚠ Important: Section “LoRaWAN Specifications” is only applicable for “Air LoRaWAN” model

2.1.5. Power supply

Feature	Description
Power consumption	2.5W ('Crowd' model) or 5W ('Air' and 'Air LoRaWAN' models)
Input	15 VDC or or Solar Panel 12V
Battery (optional)	10.05Ah (10.8V)
Power supply options	Indoor and outdoor

2.1.6. Smart Spot Core housing

Feature	Description
IP grade	IP65
Size	271 x 170 x 90 mm
Material	plastic (PC V-0; flame-resistant and self-extinguishing) Anchorage system Anti-vandalism security

Weight	1.12 kg (core housing without battery extension)
Weight	1.57 kg (core housing with battery extension)

2.1.7. Air Quality extension

Feature	Description
IP grade	IP65
Size	100 x 220 x 280 mm
Material	Aluminium
Weight	2.2 kg

2.2. Smart Spot Core socket description

Both the 'Air' and 'Crowd' models have the same arrangement of sockets on the unit, as shown in the next image.

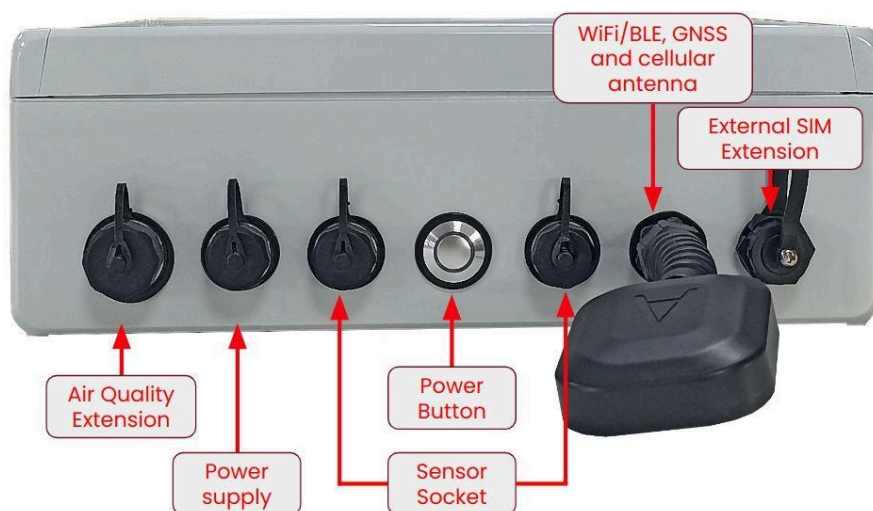


Figure: Smart Spot Crowd and Air socket description

'Air LoRaWAN' is very similar, but uses the 'Sensor Socket' closer to the 'WiFi/BLE, GNSS and Cellular Antenna' to place the LoRaWAN antenna.

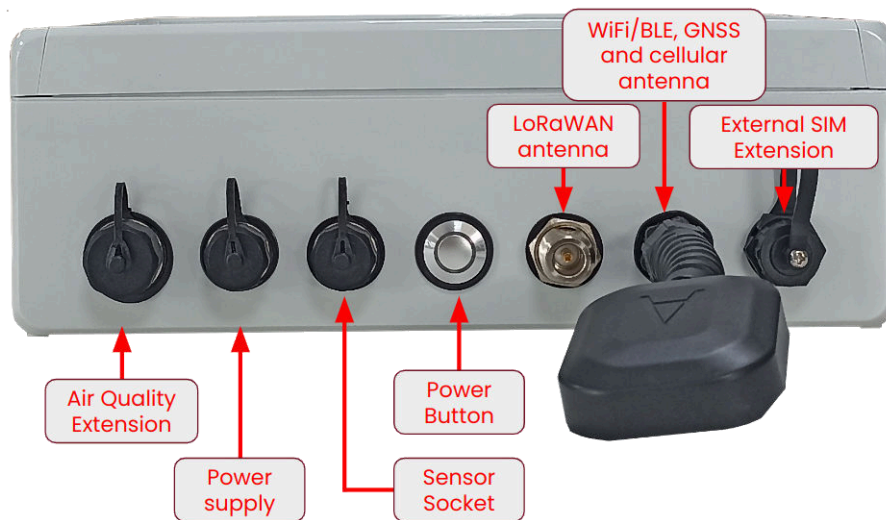


Figure: Smart Spot Air LoRaWAN socket description

2.3. Power Supply sources

At least one of the following (indoor, outdoor or solar panel) options must be selected to set up the Smart Spot unit. This device requires an external power supply to maintain its functionality over time.

2.3.1. Indoor

There is an **Indoor power supply** source with three different terminations depending on the regional standard:

- EU connector (type F)
- US connector (type B)
- UK connector (type G)



Figure: Indoor power supply

2.3.2. Outdoor

The **Outdoor electrical** panel includes a fuse holder, differential and a 15 VDC source.



Figure: Panel with Fuse holder, Differential and a 15 VDC source

2.3.3. Solar Panel

A solar panel extension (SPT-EXT-SP-30W) is available to allow the device to operate completely off-grid.

It's recommended to combine this extension with the sleeper modes such as 'LoRaWAN Sleeper' (2) or 'M2M Sleeper' (3) to get maximum performance.

The extension includes a bracket that allows the solar panel to be mounted on poles up to 210mm in diameter, as well as on walls.

Feature	Description
Size	388 x 674 x 25mm
Number of cells	36(4x9)
Cell Type	MONO CELL
Maximum Power(Wp)	30W
Maximum Power Voltage(Vm)	18.3V
Current(I _m)	1.64A
Product Certificate	TUV(IEC 61215,IEC 61730),CE,ROHS



Figure: 30W Solar Panel extension

2.3.4. Power Supply scenarios

The above power sources, combined with the optional battery extension, provide up to four different scenarios to suit any installation requirement.

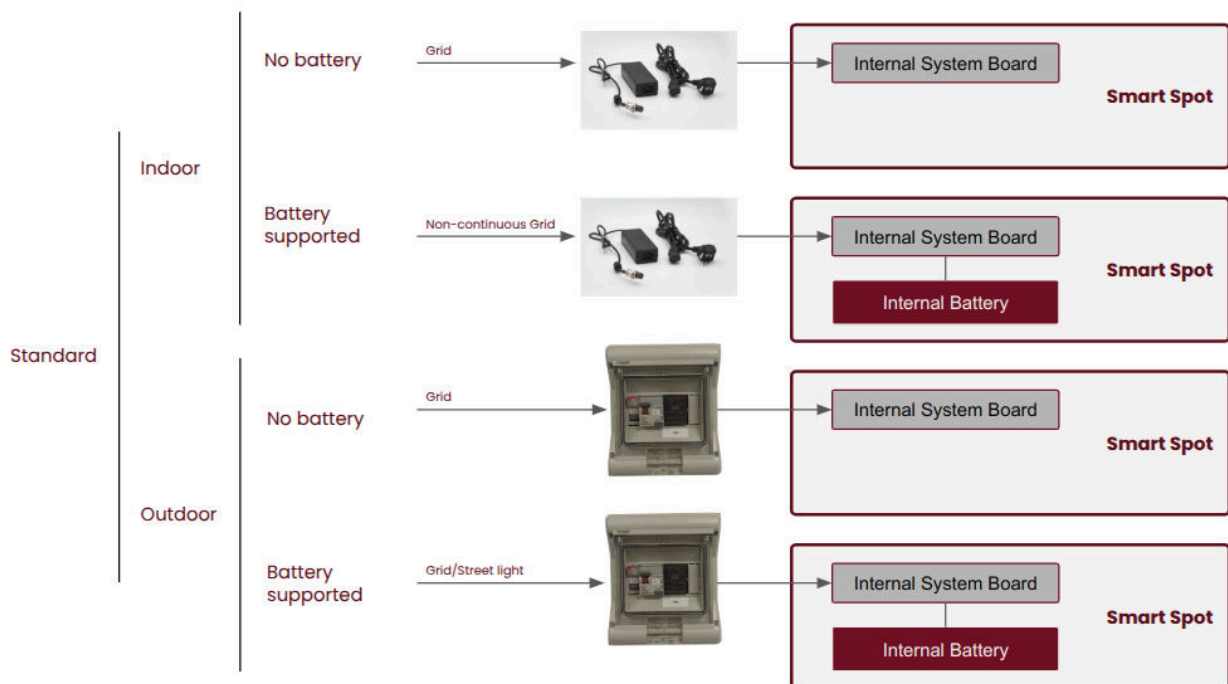


Figure: Power supply combinations

2.4. Extreme conditions battery

It is important to remember that standard batteries from any manufacturer are not designed to be charged below 0° or above 45° for safety reasons, and it is possible to find specific hot and cold conditions that may exist in certain locations for certain scenarios, such as:

- Use of the solar panel during the day in hot places (deserts, ...)
- Use of battery backed street lighting in cold locations (mountains, ...)

The extreme conditions batteries have identical capabilities regarding voltage and capacity, but these allow them to work on these special locations avoiding any kind of problem related to these special scenarios.

Feature	Description
Lower charge temperature	-30°
Maximum charge temperature	85°

2.5. On/Off button

The On/Off button is used to switch the unit on and off. When the button is pressed, the device will start up and after a few seconds a blue LED will light up to indicate that the device is on and ready for use.



Figure: On/Off button

2.6. External nanoSIM socket (optional)

This optional connector allows the user to install their own SIM card. If connectivity is purchased to Libelium directly, then there is no need for an external SIM socket as the SIM card is installed internally.

2.6.1. SIM installation

The following pictures show how to install a new SIM card if required:

Step 1: Open the SIM socket



Figure: SIM socket opening procedure

Step 2: Remove the current installed SIM card

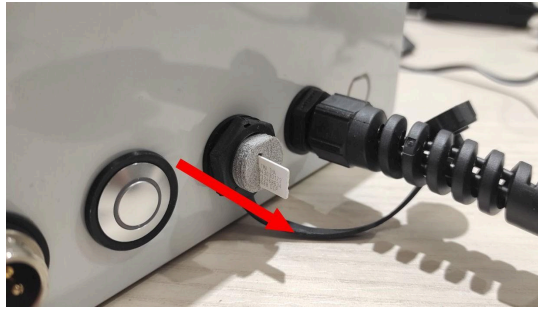


Figure: SIM removal procedure

Step 3: Install the new SIM card



Figure: SIM insertion procedure

Step 4: Close the socket protection



Figure: SIM socket closing procedure

2.7. Smart Spot extensions

2.7.1. Extension availability per model

This section explains all the extensions that can be installed in the Smart Spot core housing, depending on the device model.

2.7.1.1. "Crowd" model

Extension	Combo 1	Combo 2	Combo 3	Combo 4	Combo 5	Combo 6
Davis Pro 2 sensor	No	Yes	No	No	No	No
Davis Pro 2 plus sensor	No	No	Yes	No	No	No
Temperature and Humidity sensor	Yes	Yes	Yes	Yes	Yes	No
Wind anemometer & direction sensor	No	No	No	Yes	No	No
Atmospheric pressure sensor	No	No	No	No	No	No
AWDTHP sensor	No	No	No	No	Yes	No
Noise Level sensor	No	No	No	No	No	Yes

2.7.1.2. 'Air' model

Extension	Combo 1	Combo 2	Combo 3	Combo 4	Combo 5
Davis Pro 2 sensor	No	Yes	No	No	No
Davis Pro 2 plus sensor	No	No	Yes	No	No
Temperature and Humidity sensor	Yes	Yes	Yes	Yes	Yes
Wind anemometer & direction sensor	No	No	No	Yes	No
Atmospheric pressure sensor	Yes	Yes	Yes	Yes	Yes
AWDTHP sensor	No	No	No	No	Yes

2.7.1.3. 'Air' LoRaWAN' model

Extension	Combo 1
Temperature and Humidity sensor	Yes
Atmospheric pressure sensor	Yes

2.7.2. Temperature and Humidity sensor

The temperature and humidity sensor is fitted with a sunshield to improve sensor readings.



Figure: Temperature & Humidity sensor

Humidity specifications:

Feature	Description
Accuracy tolerance	+/- 1.5 %RH
Resolution	0.01 %RH
Range	0 to 100 %RH
Long-term drift	<0.25 %RH/year

Temperature specifications:

Feature	Description
Accuracy tolerance	+/- 0.1 °C
Resolution	0.01 °C
Range	-40 to 125 °C
Long-term drift	<0.03 °C/year

2.7.3. Davis Pro2 and Davis Pro2 Plus sensors

Davis Pro2 weather station includes next sensors:

- Rain collector
- Anemometer
- Temperature and humidity sensors

Davis Pro2 Plus weather station includes next sensors:

- Rain collector
- Anemometer
- Temperature and humidity sensors
- UV Sensor

- Solar Radiation Sensor

General specifications:

Parameter	Description
Operating Temp.	40° to +150°F (-40° to +65°C)
Non-operating Temp.	-40° to +158°F (-40° to +70°C)
Current Draw	5 mA (average) at 4 to 6 VDC for ISS
Connectors	Modular RJ11
Cable Type	4-conductor, 26 AWG
Cable Length, Anemom.	40' (12 m) (included); 240' (73 m) (maximum recommended)
Wind Speed Sensor	Solid state magnetic sensor
Wind Direction Sensor	Wind vane with potentiometer
Rain Collector Type	Tipping spoon, 0.01" or 0.2 mm per tip, 33.2 in 2 (214 cm 2) collection area

Barometric Pressure specifications:

Parameter	Description
Resolution and Units	0.01" Hg, 0.1 mm Hg, 0.1 hPa/mb
Range	16.00" to 32.50" Hg, 410 to 820 mm Hg, 540 to 1100 hPa/mb
Elevation Range	-999' to +15,000' (-600 m to 4570 m)
Accuracy	±0.03" Hg (±0.8 mm Hg, ±1.0 hPa/mb)
Sea-Level Reduction Equation Used	United States Method employed prior to use of current "R Factor" method

Dewpoint specifications:

Parameter	Description
Resolution and Units	1°F or 1°C (user-selectable) °C is converted from °F rounded to nearest
Range	105° to +130°F (76° to +54°C)
Accuracy	±2°F (±1°C) (typical)

Evapotranspiration (calculated, requires solar radiation sensor) specifications:

Parameter	Description
Resolution and Units	1°F or 1°C (user-selectable) °C is converted from °F rounded to nearest
Range	Daily to 32.67" (832.2 mm); Monthly & Yearly to 199.99" (1999.9 mm)
Accuracy	Greater of 0.01" (0.25 mm) or ±5%

Rainfall specifications:

Parameter	Description
Resolution and Units	0.01" or 0.2 mm (user-selectable)
Daily/Storm Rainfall Range	0 to 99.99" (0 to 999.8 mm)
Monthly/Yearly/Total Rainfall Range	0 to 199.99" (0 to 6553 mm)
Accuracy	For rain rates up to 10"/hr (250 mm/hr)

Rain Rate specifications:

Parameter	Description
Resolution and Units	0.01" or 0.1 mm (user-selectable)
Range	0, 0.4"/hr (1 mm/hr) 0 to 30"/hr (0 to 762 mm/hr)
Accuracy	±5% for rain rates up to 10"/hr (250 mm/hr)

Solar Radiation specifications:

Parameter	Description
Resolution and Units	1 W/m ²
Range	0 to 1800 W/m ²
Accuracy	±5% of full scale (Reference: Eppley PSP at 1000 W/m ²)
Drift	up to ±2% per year

Temperature specifications:

Parameter	Description
Resolution and Units	0.1°F or 1°F or 0.1°C or 1°C
Range	-40° to +150°F (-40° to +65°C)
Accuracy	±0.5°F (±0.3°C) (typical)

Humidity specifications:

Parameter	Description
Resolution and Units	1%
Range	1 to 100% RH
Accuracy	±2%

Ultra Violet (UV) Radiation Dose (requires UV sensor) specifications:

Parameter	Description
Resolution and Units	0.1 MEDs to 19.9 MEDs; 1 MED above 19.9 MEDs
Range	0 to 199 MEDs

Accuracy	±5% of daily total
Drift	up to ±2% per year

Ultra Violet (UV) Radiation Index (requires UV sensor) specifications:

Parameter	Description
Resolution and Units	0.1 Index
Range	0 to 16 Index
Accuracy	±5% of full scale (Reference: Yankee UVB-1 at UV index 10 (Extremely High))

Wind Chill specifications:

Parameter	Description
Resolution and Units	1°F or 1°C (user-selectable); °C is converted from °F and rounded to the nearest 1°C
Range	-110° to +135°F (-79° to +57°C)
Accuracy	±2°F (±1°C) (typical)
Update Interval	10 to 12 seconds
Source	United States National Weather Service NWS)/NOAA
Equation Used	Osczevski (1995) (adopted by US NWS in 2001)
Variables Used	Instant Outside Temperature and 10-min. Avg. Wind Speed
Current Display Data	Instant Calculation
Current Graph Data	Instant Calculation; Hourly, Daily and Monthly Low
Historical Graph Data	Hourly, Daily and Monthly Lows
Alarm	Low Threshold from Instant Calculation

Wind Direction specifications:

Parameter	Description
Range	1 - 360°
Display Resolution	16 points (22.5°) on compass rose, 1° in numeric display
Accuracy	±3°
Update Interval	2.5 to 3 seconds
Current Graph Data	Instant Reading (user adjustable); 10-min. Dominant; Hourly, Daily, Monthly Dominant
Historical Graph Data	Past 6 10-min. Dominants on compass rose only; Hourly, Daily, Monthly Dominants

Wind Speed specifications:

Parameter	Description
Resolution and Units	1 mph, 1 km/h, 0.4 m/s, or 1 knot (user-selectable) Measured in mph

Range	0 to 200 mph, 0 to 173 knots, 0 to 89 m/s, 0 to 322 km/h
Accuracy	± 2 mph (2 kts, 3.2 km/h, 0.9 m/s) or $\pm 5\%$, whichever is greater



Figure: Davis Vantage Pro2 (left) and Davis Vantage Pro 2 Plus (right) models

2.7.4. Atmospheric pressure sensor

Parameter	Description
Resolution	0.03 Pa
Absolute accuracy	± 0.5 hPa
Operating range	300 to 1250 hPa
Temperature range	-40 to 125 °C
Humidity range	0 to 100 %
Long-term stability	± 0.33 hPa (12 months)

2.7.5. Wind anemometer and direction sensor

Parameter	Description
Wind speed accuracy	$\pm(0.3 + 0.03V)$ m/s (V stands for real-time wind speed value)
Wind speed resolution	0.1 m/s
Wind speed range	0 to 70 m/s
Wind direction accuracy	$\pm 3^\circ$
Wind direction resolution	1°
Wind direction range	0 to 360°
Starting wind speed	<0.5 m/s
Power supply	9 to 30 V

Max. power consumption	0.9 W
Output signal	RS485
Line length	3 metres



Figure: Wind anemometer and direction sensor

2.7.6. Wind speed/direction, temperature, humidity and pressure sensor (AWDTHP)

AWDTHP includes the following sensors:

- Atmospheric temperature
- Atmospheric humidity
- Atmospheric pressure
- Wind speed
- Wind direction

General specifications:

Parameter	Description
Supply	12-24 VDC, solar power
Output	RS485 (Modbus)
Power Consumption	< 1W @ 12V
Ingress Protection	IP65
Main Material	ASA + engineering plastics

Temperature specifications:

Parameter	Description
Principle	Energy gap method

Resolution and Units	0.1°C
Range	-40° to +60°C
Accuracy	±0.3°C (25°C)

Humidity specifications:

Parameter	Description
Principle	Capacitance method
Resolution and Units	0.01%
Range	0% to 100% RH
Accuracy	±3% RH

Atmospheric Pressure specifications:

Parameter	Description
Principle	Piezoresistive
Resolution	0.1 hPa
Absolute accuracy	± 0.5 hPa (0-30°C)
Operating range	300 to 1100 hPa

Wind Speed specifications:

Parameter	Description
Principle	Ultrasonic
Resolution and Units	0.01 m/s
Range	0 to 60 m/s
Accuracy	±0.3 m/s or ±3%, whichever is greater

Wind Direction specifications:

Parameter	Description
Principle	Ultrasonic
Range	0 - 360°
Display Resolution	0.1°
Accuracy	±2°



Figure: Wind speed/direction, temperature, humidity and pressure sensor

2.7.7. Noise level sensor

Specifications:

Characteristic	Description
SPL Working Range	40 to 115 dB
Frequency Weighting	IEC 61672-1 A Filter
Time Weighting	IEC 61672-1 Slow (S) and Fast (F)
Power Consumption	40mA Active
Supply Voltage (nominal)	5V
Supply Voltage (limits)	3.6V to 6V
RoHS	Yes
Functions provided	LAmin, LAmax, LAeq, LA1, LA10, LA50, LA90, LA99

Measurements provided by the sensor:

Function	Description
LAmin	Minimum value of the measurement in the currently time window
LAmax	Maximum value of the measurement in the currently time window
LAeq	Equivalent continuous is the average of sound pressure level (acoustic energy over the time window)
LA1	Value of noise level exceeded for 1% of the measurement period with 'A' frequency weighting
LA10	Value of noise level exceeded for 10% of the measurement period with 'A' frequency weighting

LA50	Value of noise level exceeded for 50% of the measurement period with 'A' frequency weighting
LA90	Value of noise level exceeded for 90% of the measurement period with 'A' frequency weighting
LA99	Value of noise level exceeded for 99% of the measurement period with 'A' frequency weighting



Figure: Noise Level sensor

3. Models

3.1. Smart Spot 'Crowd'

The **Smart Spot Crowd** model device MUST be configured with one of the following reporting modes:

- Count reporting
- Count + MAC/SSID reporting



Figure: Smart Spot Crowd

3.1.1. Count reporting

The main purpose of this mode is to detect, temporarily accumulate and **report the number of devices detected** in the last few minutes to track the flow of devices in the location where the device is installed.

The accumulation is done in different configurable time intervals, which by default are 1 min, 5 min and 10 min. The reporting interval is the smallest of the configurable time intervals.

The service configuration is done (via Homard) in the LWM2M object 'LBL: Crowd Monitor'. It is possible to configure parameters for detection, among them:

- **Lower counter interval:** Lower accumulation time interval, will be used as interval for data reporting. (minimum 1 min and always less than medium and higher)
- **Medium counter interval:** Intermediate accumulation time interval. (this parameter must be always greater than lower and less than higher).
- **Higher counter interval:** Long accumulation time interval. (This parameter is limited to 10 minutes and must always be greater than lower and medium).
- **WiFi RSSI Detection Threshold:** Detection limit setting (subject to the emission force of the detectable device).
- **BLE RSSI Detection Threshold:** Detection limit setting (subject to the emission strength of the detectable device).
- **MAC filter:** Type of MACs to detect (random, non-random, all)

The next image shows the different counting intervals provided by Smart Spot Crowd. The number of devices detected within each interval (e.g. 1min, 5min, 10min) is shown in a line graph:

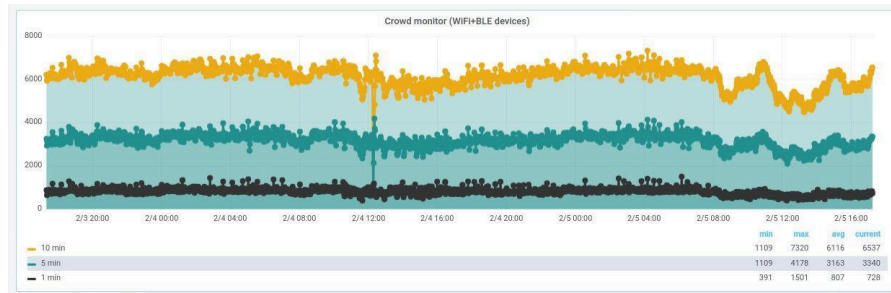


Figure: Count reporting

3.1.2. Count + MAC/SSID reporting

In this mode, in addition to counting the number of devices, the Smart Spot adds the ability to process and report **extra information extracted** from detected devices. For each device detected, a report is sent in which you will find the MAC identifier obfuscated by the **SHA1 hash algorithm** (by default). The obfuscation format is intended for privacy purposes.

The mode configuration is done (via Homard) in the LwM2M object 'LBL: Crowd Monitor'. It is possible to configure additional parameters:

- **Report MAC Mode:** Selection of identifier type (WiFi/BLE) and report protocol (LwM2M/MQTT/SENTILO).
- **Report MAC Format:** Information report format: RAW or SHA1 (default).
- **Report MAC Type:** Configuration of resources used to report identifiers via LwM2M

Refer to **AN20: Crowd Monitoring technology and configuration via Homard** for extended information about the technology and available configurations.

3.2. Smart Spot 'Air' and 'Air LoRaWAN'

The **Smart Spot Air** and **Smart Spot Air LoRaWAN** models come along with a secondary housing with gas sensors and other list of optional sensor extensions.



Figure: Smart Spot Air Quality

3.2.1. Gas sensors

Note: These are optional sensors

Please refer to the following table for the list of **gas sensors** available. All pollutants are divided into different application areas, taking into account the cross sensitivity between some sensors.

	Smart Cities	Cities near Industry	Industry	Farming
NO ₂	Yes	Yes	Yes	No
O ₃	Yes	Yes	Yes	No
CO	Yes	Yes	Yes	Yes
SO ₂	Yes	Yes	Yes	No
NO	Yes	Yes	Yes	Yes
H ₂ S	No	Yes	Yes	Yes
NH ₃	No	No	Yes	Yes

The **cross sensitivity** between the gas sensors is:

	NO ₂	O ₃	CO	SO ₂	NO	H ₂ S	NH ₃
with NO ₂	100%	100%	0%	-120%	7%	-20%	0%
with O ₃	0%	100%	0%	-120%	0%	0%	0%
with CO	0%	3%	100%	3%	0%	0%	0%
with SO ₂	0%	3%	0%	100%	0%	15%	45%
with NO	5%	5%	0%	4%	100%	0%	0%

with H2S	-80%	-80%	0%	0%	20%	100%	50%
with NH3	0%	0%	0%	0%	0%	0%	100%

3.2.1.1. NO2 sensor

Specification	Description
Range	0-5 ppm
Accuracy	5 ppb (± 2 standard deviations)
Resolution	1 ppb
Overgas limit	50 ppm
Response time	< 60 s (t90 from zero to 2ppm NO2)
Operating life	> 24 months
Temperature range	-30 to 40 °C
Humidity range	15 to 85 %RH continuous
Pressure range	80 to 120 kPa

3.2.1.2. O3 sensor

Specification	Description
Range	0-5 ppm
Accuracy	15 ppb (± 2 standard deviations)
Resolution	1 ppb
Overgas limit	50 ppm
Response time	< 45 s (t90 from zero to 1 ppm O3)
Operating life	> 24 months
Temperature range	-30 to 40 °C
Humidity range	15 to 85 %RH continuous
Pressure range	80 to 120 kPa

3.2.1.3. CO sensor

Specification	Description
Range	0-10 ppm
Accuracy	20 ppb (± 2 standard deviations)
Resolution	1 ppb
Overgas limit	2000 ppm
Response time	< 20 s (t90 from zero to 10 ppm CO)
Operating life	> 24 months
Temperature range	-30 to 50 °C
Humidity range	15 to 90 %RH continuous

Pressure range	80 to 120 kPa
----------------	---------------

3.2.1.4. SO₂ sensor

Specification	Description
Range	0-5 ppm
Accuracy	15 ppb (± 2 standard deviations)
Resolution	1 ppb
Overgas limit	100 ppm
Response time	< 20 s (t ₉₀ from zero to 2 ppm SO ₂)
Operating life	> 24 months
Temperature range	-30 to 50 °C
Humidity range	15 to 90 %RH continuous
Pressure range	80 to 120 kPa

3.2.1.5. NO sensor

Specification	Description
Range	0-5 ppm
Accuracy	15 ppb (± 2 standard deviations)
Resolution	1 ppb
Overgas limit	50 ppm
Response time	< 25 s (t ₉₀ from zero to 2 ppm NO)
Operating life	> 24 months
Temperature range	-30 to 50 °C
Humidity range	15 to 85 %RH continuous
Pressure range	80 to 120 kPa

3.2.1.6. H₂S sensor

Specification	Description
Range	0-1 ppm
Accuracy	5 ppb (± 2 standard deviations)
Resolution	1 ppb
Overgas limit	100 ppm
Response time	< 45 s (t ₉₀ from zero to 2 ppm H ₂ S)
Operating life	> 24 months
Temperature range	-30 to 50 °C
Humidity range	15 to 90 %RH continuous
Pressure range	80 to 120 kPa

3.2.1.7. NH3 sensor

Specification	Description
Range	0-20 ppm
Accuracy	0.5 ppm
Resolution	1 ppb
Overgas limit	200 ppm
Response time	< 60 s
Operating life	> 24 months
Temperature range	-30 to 50 °C
Humidity range	15 to 90 %RH (no condensation)
Pressure range	80 to 120 kPa

3.2.1.8. VOC sensor

Specification	Description
Range	0-40 ppm
Resolution	1 ppb
Response time	< 8 s
Operating life	> 5 years
Temperature range	-40 to 55 °C
Humidity range	15 to 90 %RH continuous
Pressure range	80 to 120 kPa

3.2.2. CO2 sensor

Characteristic	Description
Measurement range	400 - 10000 ppm
Accuracy (1)	±30 ppm
Repeatability (2)	± 10 ppm
Response time (τ63%)	20 s
Temperature operating conditions	0 – 50°C
Humidity operating conditions	0 – 95 %RH (Non-condensing)
Lifetime	15 years

(1): Deviation to a highprecision reference in the calibrated range (400 – 10'000 ppm) of the SCD30. Accuracy is fulfilled by > 90% of the sensors after calibration. Rough handling, hiping and soldering reduces the accuracy of the sensor. Full accuracy is restored with FRC or ASC

recalibration features. Accuracy is based on tests with gas mixtures having a tolerance of $\pm 1.5\%$.

(2): RMS error of consecutive measurements at constant conditions. Repeatability is fulfilled by $> 90\%$ of the sensors

3.2.3. Particulate Matter sensor

Particulate matter consists of small solid or liquid particles suspended in the air. The source of these particles can be industrial activity, exhaust fumes from diesel engines, building heating, pollen, etc. These tiny particles enter our bodies when we breathe. High concentrations of particulate matter can be harmful to humans and animals, leading to respiratory and coronary diseases and even lung cancer. This is why it is a key parameter in the Air Quality Index.

Particles are classified as follows:

- **PM1:** Mass (in μg) of all particles smaller than $1 \mu\text{m}$, in 1 m^3 .
- **PM2.5:** Mass (in μg) of all particles smaller than $2.5 \mu\text{m}$, in 1 m^3 .
- **PM4:** Mass (in μg) of all particles smaller than $4 \mu\text{m}$, in 1 m^3 .
- **PM10:** Mass (in μg) of all particles smaller than $10 \mu\text{m}$, in 1 m^3 .
- **TSP:** Mass (in μg) of all Total Suspended Particulates in 1 m^3 .
- **TPC:** Total Particle Counter in 1 cm^3 .

⚠ Note: High humidity or fog may affect the sensor's readings. Particles may be swollen, coated with water or agglutinated.

Specifications:

Characteristic	Description
Particle range*	0.35 to $40 \mu\text{m}$
Max particle count rate	10000 Particles/second
Measurements	PM1, PM2.5, PM4, PM10, TSP and TPC
Resolution	$0.1 \mu\text{m}/\text{m}^3$ (count/ cm^3 for TPC)
Temperature range	-10 to $50 \text{ }^\circ\text{C}$
Humidity range	0 to 95 %RH (non-condensing)
Lifecycle	24 months

(*) Based on 100% detection efficiency at $0.35 \mu\text{m}$, 50% at $0.3 \mu\text{m}$

3.2.4. Noise level sensor

Specifications:

Characteristic	Description
SPL Working Range	40 to 115 dB
Frequency Weighting	IEC 61672-1 A Filter
Time Weighting	IEC 61672-1 Slow (S) and Fast (F)
Power Consumption	40mA Active
Supply Voltage (nominal)	5V
Supply Voltage (limits)	3.6V to 6V
RoHS	Yes
Functions provided	LAmin, LAmax, LAeq, LA1, LA10, LA50, LA90, LA99

Measurements provided by the sensor:

Function	Description
LAmin	Minimum value of the measurement in the currently time window
LAmax	Maximum value of the measurement in the currently time window
LAeq	Equivalent continuous is the average of sound pressure level (acoustic energy over the time window)
LA1	Value of noise level exceeded for 1% of the measurement period with 'A' frequency weighting
LA10	Value of noise level exceeded for 10% of the measurement period with 'A' frequency weighting
LA50	Value of noise level exceeded for 50% of the measurement period with 'A' frequency weighting
LA90	Value of noise level exceeded for 90% of the measurement period with 'A' frequency weighting
LA99	Value of noise level exceeded for 99% of the measurement period with 'A' frequency weighting

4. Hardware setup and installation

4.1. General recommendations

- Can be mounted on wall, pole or street light
- The recommended installation height is between 3 and 3.5 metres to meet various requirements (vandalism prevention, noise position, etc.).
- If the device includes Air Quality
 - It is recommended that the units are installed in open and clean areas, but accessible to facilitate maintenance.
 - Avoid nearby transformers and antennas that generate strong electromagnetic fields that can interfere with gas sensors.
 - Exclude certain areas in which it is allowed to smoke or there are smoke extractors.
 - Consider the type of air quality measurement you are interested in. There are two main types of air quality measurement:
 - * Emission (direct pollution): The device must be placed near the source of pollution (near a road, ...), for example 5 metres from the source of pollution.
 - * Emission (background pollution): The device must be installed in a place where the pollution is dispersed, e.g. open spaces at a distance of 50m from the pollution source (parks, gardens, etc.).
- If the device includes noise level sensor
 - The ISO 19962 2009 standard recommends installing the unit at a distance of 2 metres from a wall. If this isn't possible, the Smart Spot allows you to configure a correction parameter for wall installation (3dB), as described on the standard, to avoid the reflected noise.

4.2. Power supply installation

Regardless of the model selected, the power supply is connected to the Smart Spot via a 2-pin connector, located as previously illustrated in the 'Smart Spot Core socket description' section.

4.2.1. Indoor power supply

- Connect the 2-pin connector to the Smart Spot
- Connect the power supply to the Schuko plug.



Figure: Power supply installation

4.2.2. Outdoor power supply

- Connect the L/N wires to the fuse (the markings in the fuse indicate where to connect).
- Connect the 2-pin cable to the transformer
 - Connect the blue cable to -V (corresponds to N line)
 - Connect the brown cable to +V (corresponds to L line)

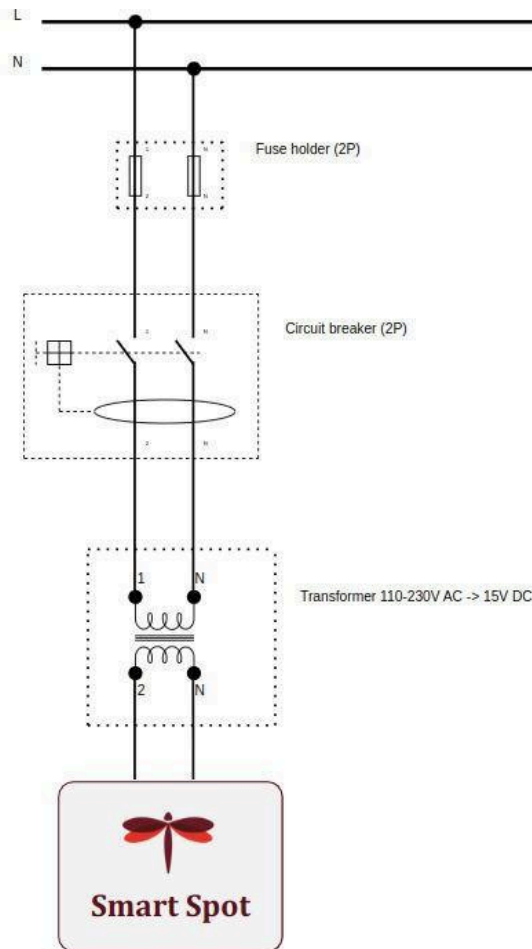


Figure: Electrical schema for outdoor power supply

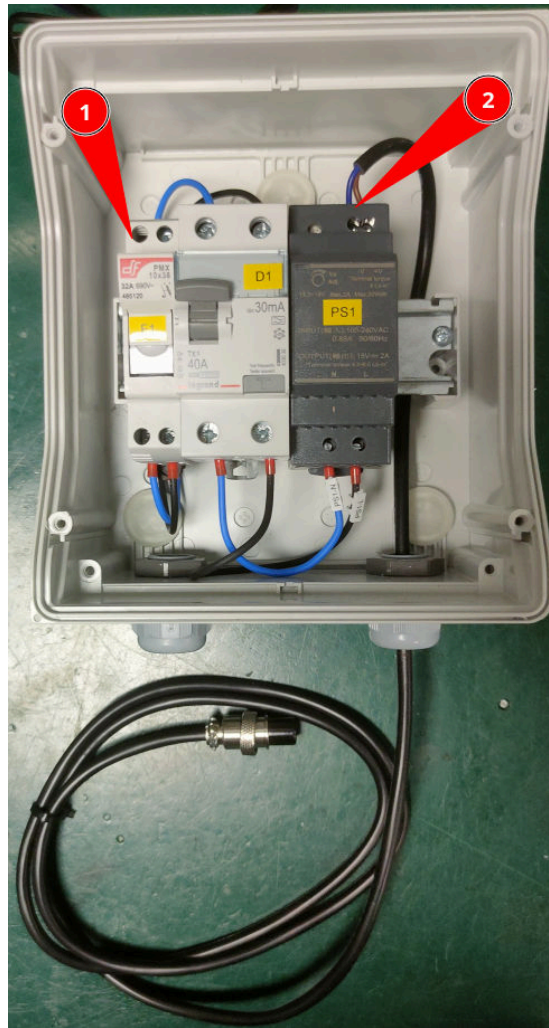


Figure: Outdoor power supply connections: 1) L/N input; 2) 15VDC output

4.2.3. Solar Panel

The ideal orientation of solar panels is based on pointing them towards the equator (south in the northern hemisphere and north in the southern hemisphere) and adjusting the tilt angle according to the latitude of the installation site.

- Northern hemisphere: Solar panels should face south
- Southern hemisphere: Solar panels should face north

The optimum tilt angle of a solar panel depends on the latitude of the installation site. A tilt angle equal to the latitude of the site is usually a good approximation. For example, if the installation is in a location with a latitude of 30 degrees, the panels should be tilted at 30 degrees to the horizontal plane.

Other important considerations when installing the solar panel are:

- Shading: Ensure that the panels are not shaded during peak hours of sunlight. This includes avoiding shading from trees, buildings or other obstructions.
- Ventilation: Leave enough space between the panels and the mounting surface to allow for good ventilation, which helps prevent overheating and improves panel efficiency.

Once you are sure of the location and position of the solar panel, follow the instructions below:

- Attach the solar panel to the bracket (1)
- Fasten the bracket to the wall using a screw or to the pole using clamps. (2)
- Ensure that the extension is correctly and securely attached.

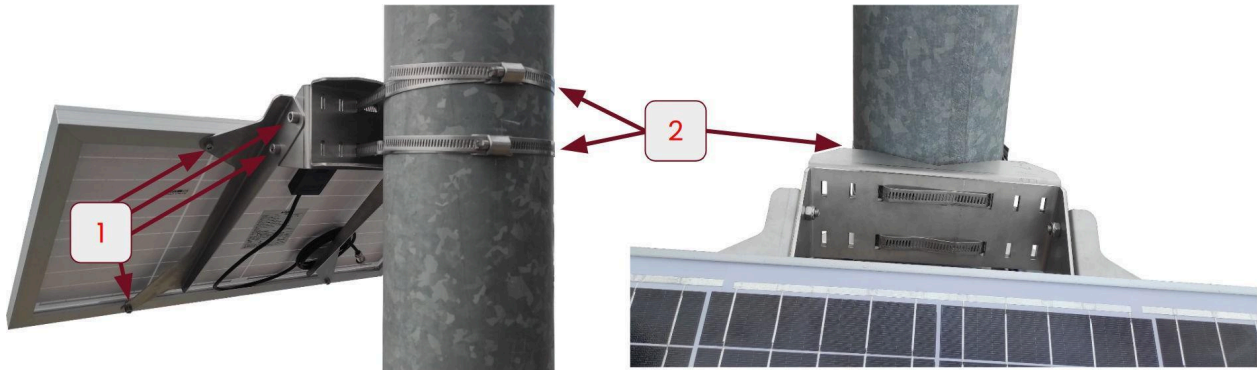


Figure: Solar panel pole installation guide

Note: If possible, it's good to use the solar panel to protect the main Smart Spot enclosure from direct sunlight to protect the system from high temperatures. For Air Quality models, it's better to protect the Air Quality case to give the system a more stable temperature.

4.3. Sensor socket installation

Depending on the Smart Spot model, you will find different sensor socket arrangements on the core housing. The 'Crowd' and 'Air' models allow the connection of two different external sensor extensions, while 'Air LoRaWAN' occupies one of the sensor sockets with the LoRaWAN antenna.

4.3.1. Smart Spot 'Crowd' and 'Air'

The list of external sensors/modules to be connected (not all at once) is as follows:

- Temperature and Humidity sensor
- Davis Pro2 sensor
- Davis Pro2 Plus sensor
- Anemometer & wind direction sensor
- AWDTHP sensor
- Noise Sensor (external, just for the 'Crowd' model)

These must be connected according to their combination.



Figure: Combo 1: Temperature & Humidity sensor



Figure: Combos 2a 3a: Davis Pro2 or Davis Pro2 Plus



Figure: Combos 2b 3b: Davis sensor and Temperature and Humidity sensor



Figure: Combo 4a: Anemometer & wind direction sensor



Figure: Combo 4b: Anemometer & wind direction sensor and Temperature and Humidity sensor



Figure: Combo 5: Wind speed/direction, temperature, humidity and pressure sensor

For the 'Crowd' model, a Noise Level Sensor can be connected externally to the SmartSpot core as indicated in the following image.



Figure: Combo 6: Noise Level sensor

4.3.2. Smart Spot 'Air LoRaWAN'

The list of external sensors/modules that can be connected to the **Smart Spot Air LoRaWAN** model (not all at once) is as follows

- Temperature and Humidity sensor

These must be connected according to their combination.



Figure: Combo 1a: Temperature & Humidity sensor

4.4. Device installation

Utilities for both wall and pole mounting are provided with the Smart Spot such as illustrated on the next figures.

⚠ Note: To cover the high variability of scenarios and customers (poles, street lights of different diameters), endless clamps are offered as a separate item, enabling cost savings for customers.

4.4.1. Smart Spot Core housing



Wall mounting accessories



Pole mounting accessories

Figure: Wall and pole mounting accessories for Smart Spot core housing

4.4.1.1. Installation in pole

- Place the supports in the Core housing (1)
- Use the accessory to pass the end-less clamp and attach the device to the pole (2)

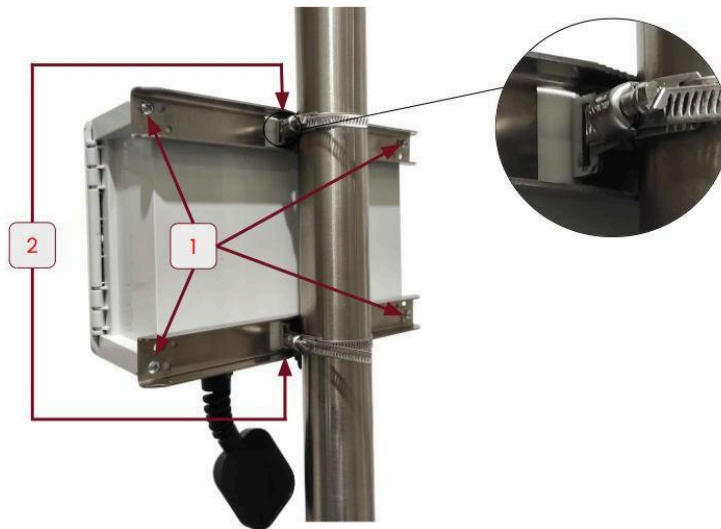


Figure: Steps to install the Core in Pole

⚠ Note: When using endless clamps, cut off the excess after tightening

4.4.2. Air Quality Extension



Figure: Wall and pole mounting accessories for Smart Spot Air Quality Extension housing

4.4.2.1. Installation in pole

- Place the supports in the Extension housing (1)
- Use the accessory to pass the end-less clamp and attach the device to the pole (2)

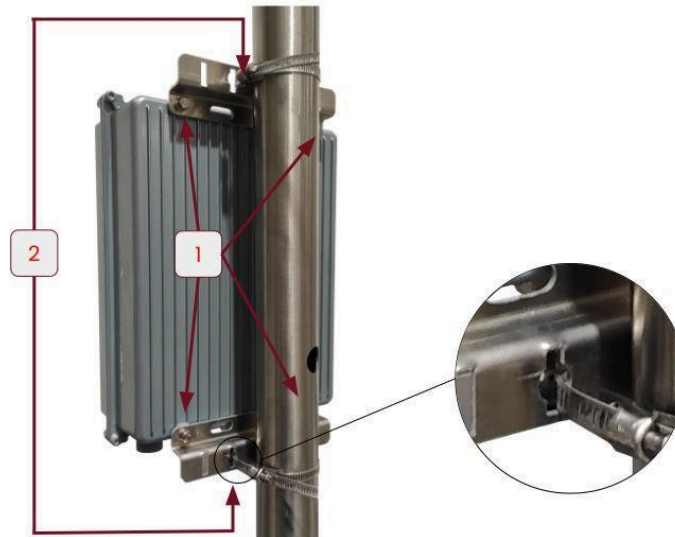


Figure: Steps to install the Air Quality Extension in Pole

⚠ Note: When using endless clamps, cut off the excess after tightening

4.5. Extensions

4.5.1. Temperature and humidity sensor

This sensor can be installed with the same clamps as the Core housing and Air Quality Extension (offered as a separate item).

- Pass the cable through the bracket (1)
- Fasten the sensor to the bracket using the wing nuts (2)
- Pass the end-less clamp through the support and attach the sensor to the pole (3)

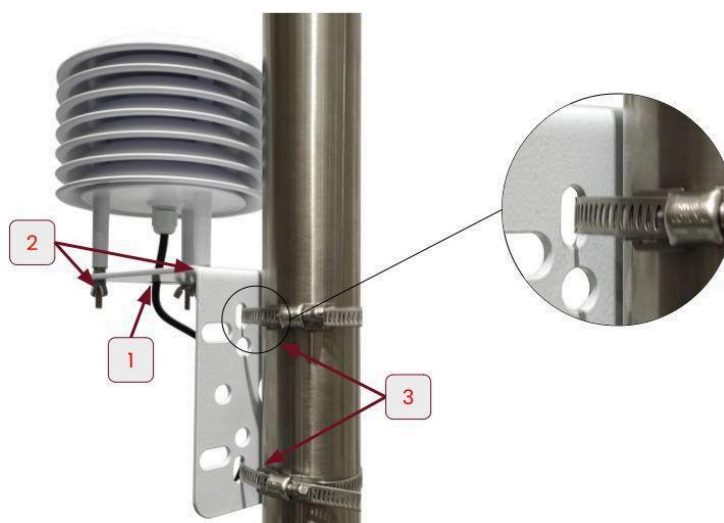


Figure: Steps to install the Temperature and humidity sensor

⚠ Note: When using endless clamps, cut off the excess after tightening

4.5.2. Wind anemometer and direction sensor

There are four ways to install this sensor, as shown in the figures below:

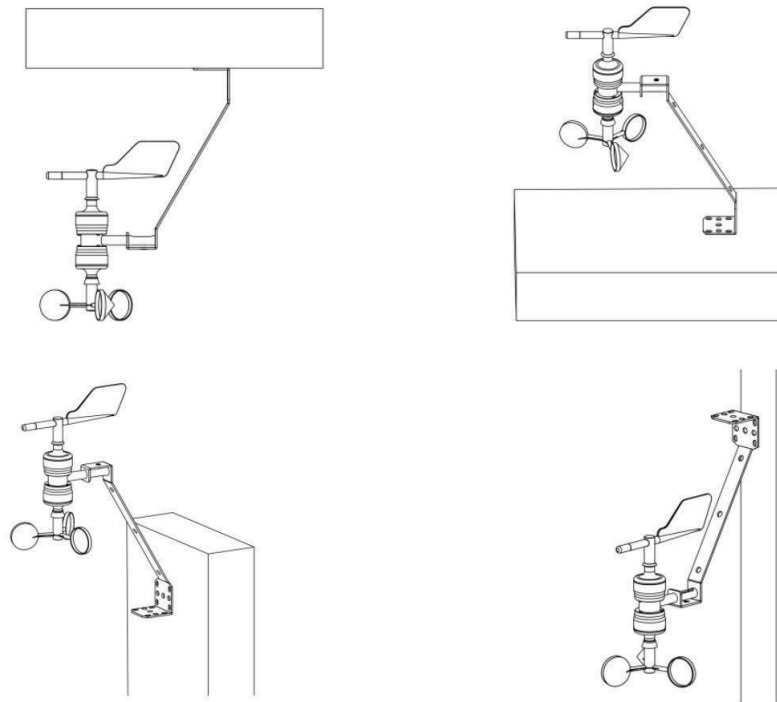


Figure: Installation of wind anemometers and direction sensors

⚠ **Important:** Note that the sensor must be installed with the wind speed sensor facing upwards. Important: Use the 'S' (South) mark on the wind direction sensor to install the sensor correctly. The S mark must point to true south. True South should not be confused with Magnetic South. The magnetic variation around the world can be quite large.

⚠ **Important:** Consider the environment to avoid placing the wind sensor under or near objects that may interfere with the measurements.

4.5.3. AWDTHP sensor

The sensor can be fastened to a pole (with maximum external diameter 50 mm and minimum internal diameter 36mm) with the included screws or by positioning flanges in the **5** position. The sensor must be installed with the **1** arrow pointing northward and with the ultrasonic probes (**2**) and thermometer shelter (**4**) free from obstructions.

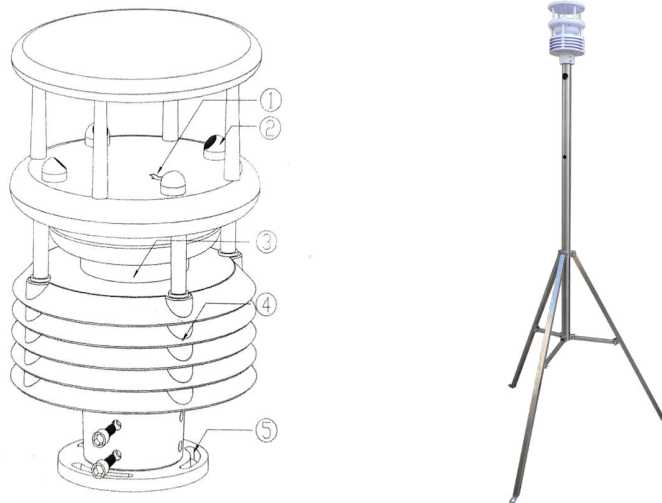


Figure: Installation of the AWDTHP sensors and example of a tripod situation

As an option, a metallic holder for the sensor is available, so the sensor can be easily attached to a pole (up to **210 mm diameter**) or to a wall. Once you are sure of the location and position of the sensor, follow these instructions:

- If the sensor is to be placed on a wall, mark the four holes by placing the bracket in the desired position, then drill the wall and insert the corresponding plugs. Finally you can screw the support to the wall.
- If the sensor is going to be placed on a lamppost or mast, insert the clamps into their housings, place the support in the desired position and finally close the clamps by firmly tightening the screws.
- Place the sensor on the support and once you are sure of the position, firmly tighten the retaining screws.



Figure: AWDTHP sensor holder installation on pole



Figure: AWDTHP sensor placed on the holder



Figure: AWDTHP sensor placed on a lamppost

4.5.4. Weather station

⚠ Important: The information in this section has been extracted from the official installation instructions found on the Weather Station box. If you need more details, please refer to the official installation instructions.

4.5.4.1. Components mounting

Follow the steps below to install the different parts of your Weather Station.

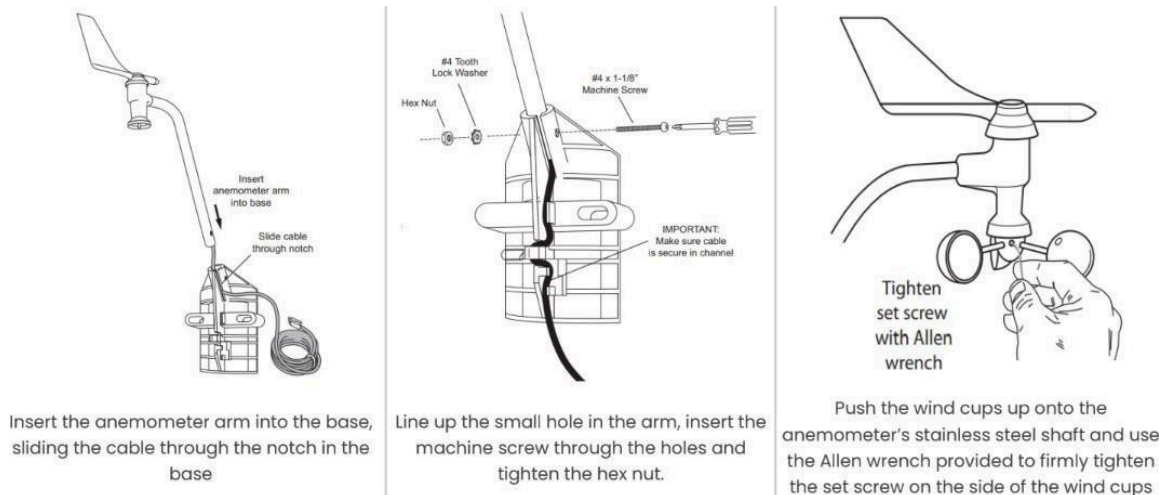


Figure: Weather station installation (instruction set 1)

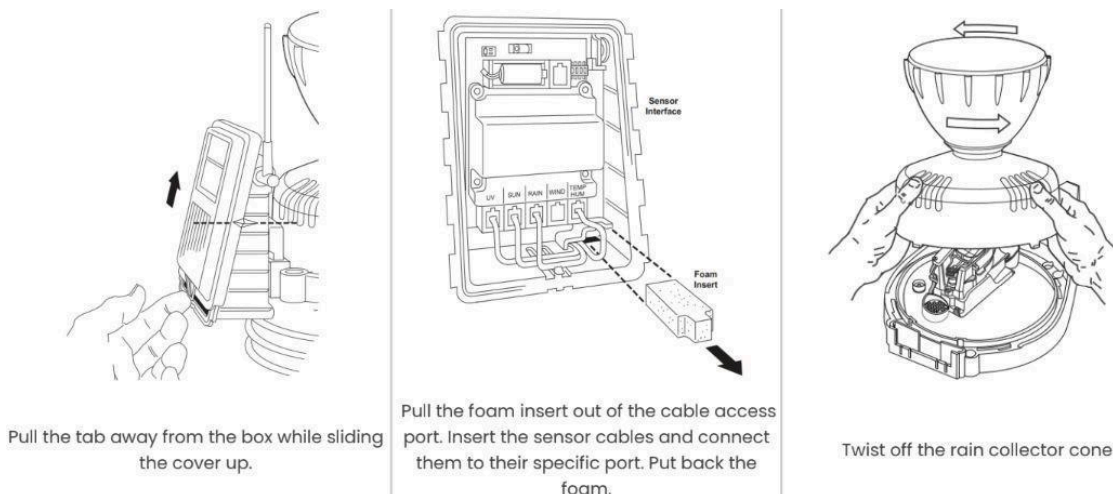


Figure: Weather station installation (instruction set 2)

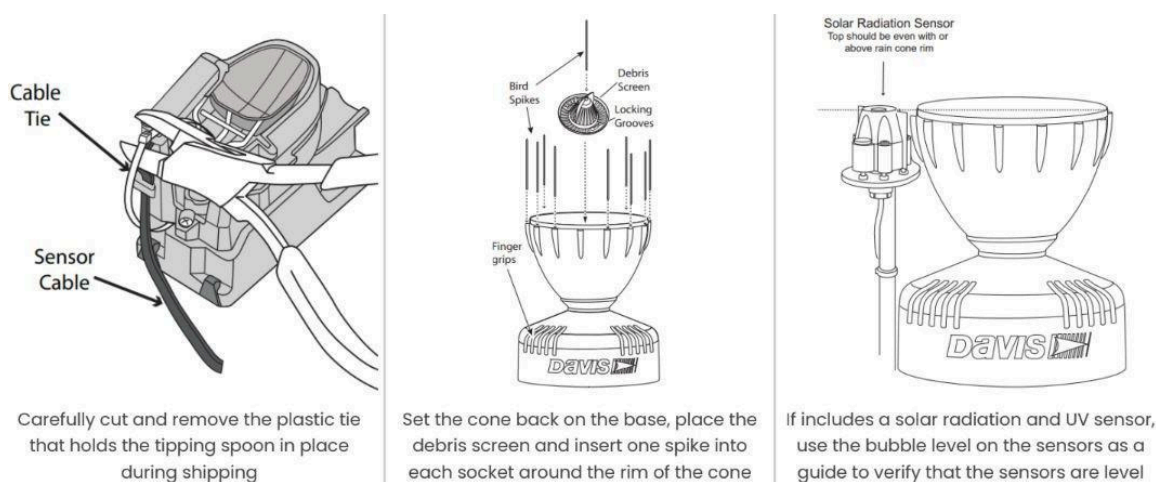


Figure: Weather station installation (instruction set 3)

⚠ Important: Remember to carefully cut and remove the plastic tie to allow rain parameters to be measured.

⚠ **Important:** The wind direction sensor arm must be aligned with true north. True North should not be confused with Magnetic North. The magnetic variation around the world can be quite large.

4.5.4.2. Pole mounting

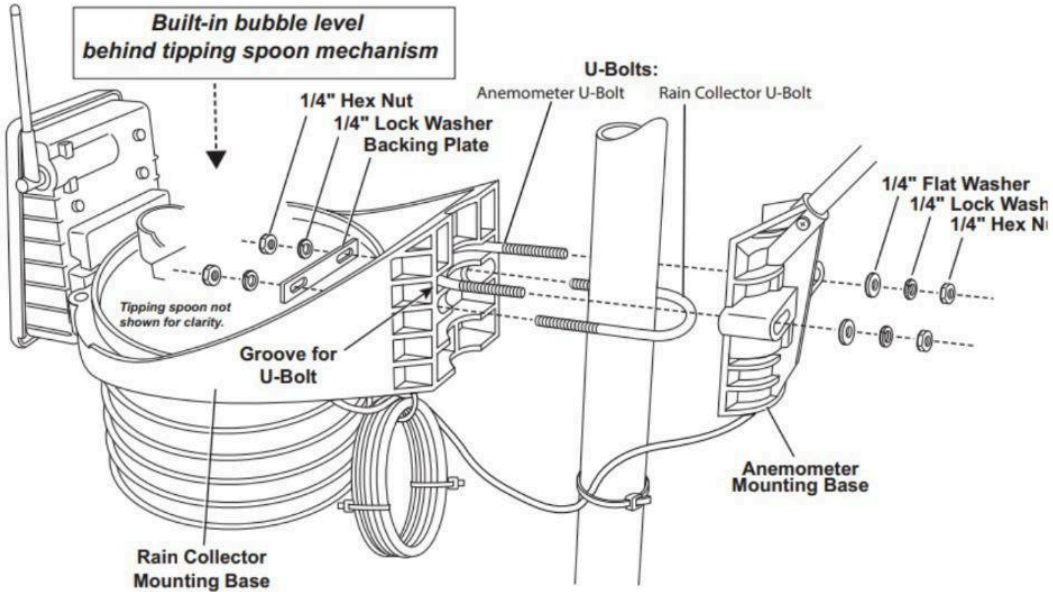


Figure: Weather station installation (Pole mounting)

4.5.4.3. Wall mounting

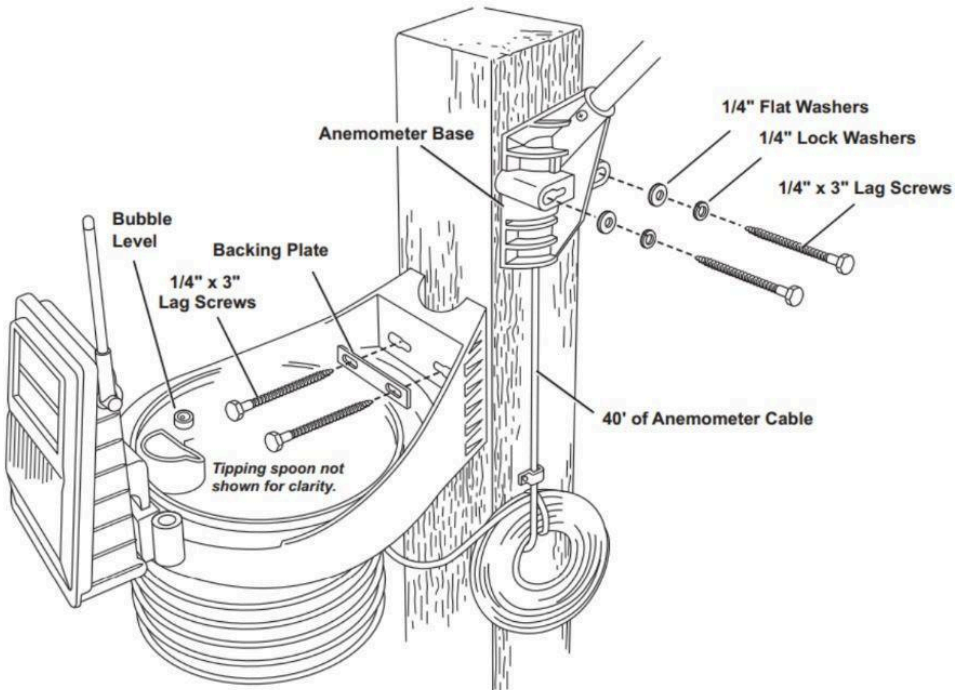


Figure: Weather station installation (Wall mounting)

4.5.4.4. Recommendations

Consider the environment to avoid placing the Weather Station under, near or between objects that may interfere with the rain, solar and wind measurements.

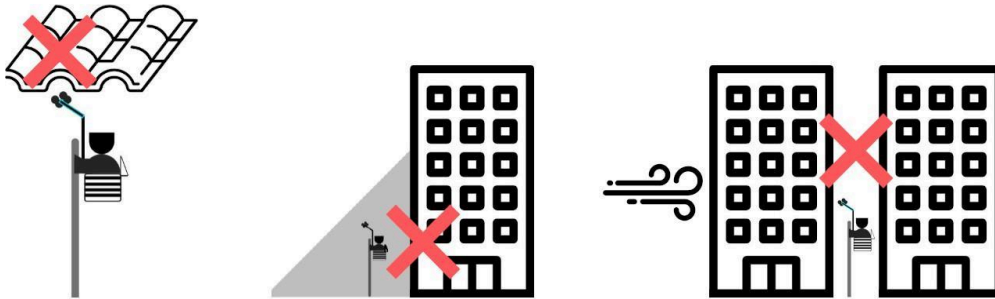


Figure: Weather station installation recommendations

4.5.5. Noise Level sensor

This sensor can be installed with the included metallic clamps, passing them through the screw holes in the sensor case (1).

- After that, fasten the sensor to the pole tightening the clamps (2)
- And cut the excess clamp (3)

Once the sensor case is secured to the pole proceed to install the microphone (4).



Figure: (1) Noise Level sensor installation recommendations (clamps placing)



Figure: (2) Noise Level sensor installation recommendations (pole installation with clamps)



Figure: (3) Noise Level sensor installation recommendations (cut excess clamp length)



Figure: (4) Noise Level sensor installation recommendations (install microphone)

4.6. Problem resolution

If you experience any problems with the device after installation, such as lack of connectivity, lack of data from sensors, etc., you can refer to the **Application Note 122: Diagnostic Guide** document for more detailed information to help you solve the problem.

5. Device setup and management

5.1. Device setup

Smart Spots have pre-configured WiFi settings so that they connect to the Internet the first time they are turned on. Therefore, once connected to the Internet, the Smart Spots can be configured with new network parameters such as APNs, WiFi access points, reporting intervals, third party integrations, ... via the Homard device management platform.

The access point (AP) that Smart Spots need to connect to for the first time can be easily set up from a smartphone hotspot or WiFi router. The default configured network is:

- SSID name: **defaultSSAP**
- Password: **defaultSSAPI234**

⚠ **Important:** The access point must be configured in the 2.4Ghz band. The 5Ghz band is not supported by the Smart Spot.

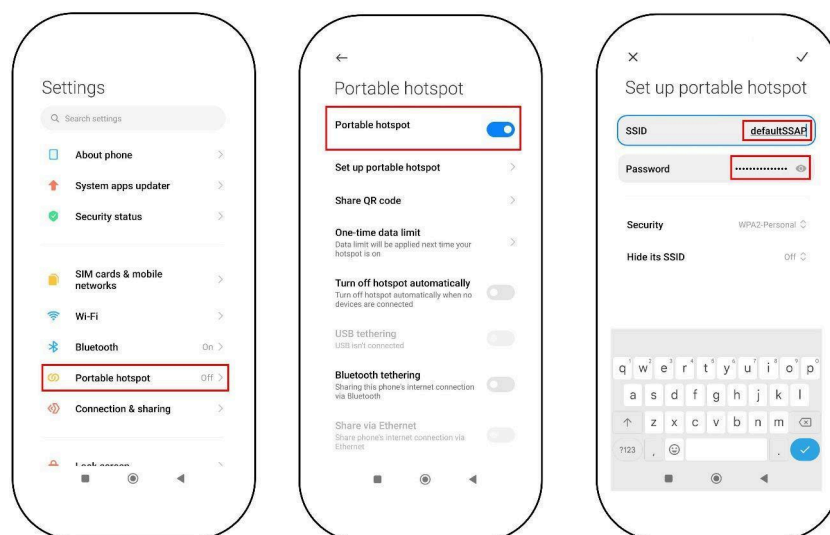


Figure: How to setup the WiFi Access Point

Once the Access Point is established, it is possible to turn on the Smart Spot device so it will search and try to connect to the WiFi network with the mentioned SSID and password. Thus the device will connect automatically to Homard platform and it will be visible in the corresponding dashboard.

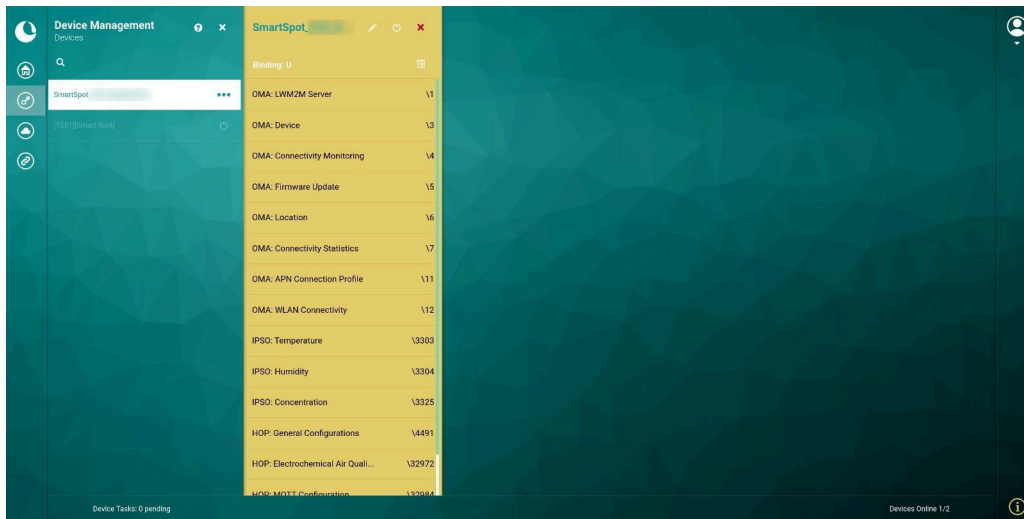


Figure: Device profile in Homard

5.2. Device management

5.2.1. LwM2M for device management

Open Mobile Alliance (OMA) **Lightweight M2M** (LwM2M) is an Open Mobile Alliance protocol for M2M for IoT device management and service enablement. It defines the application layer communication between an LwM2M Server and an LwM2M Client, providing device management functions such as bootstrap, basic operations (reboot, etc.), network configuration, and remote diagnostics, reporting or firmware upgrade, among others. It also defines a strong and standardized semantics for the endpoints or resources used for the above functions to improve the portability between systems. Protocol security is based on the use of **DTLS** or **TLS** protocols.

The organisation of the **data model** is based on a tree with a maximum depth of 4 levels. These levels are respectively: Objects, Object Instances, Resources and Resource Instances, where each level is embedded in the previous one and there may be more than one sub-level within a level (instance).



Figure: Data model hierarchy

At the 4th level, the resources correspond to the direct interaction with the device, while the previous levels provide the organisation of the data model. These resources allow from one to a combination of the following operations:

- **Read** (e.g. reading a configuration parameter).
- **Write** (e.g. write a configuration parameter such as a read interval).
- **Execute** (e.g. perform a remote device reboot).
- **Observe** (e.g., receive each new measurement automatically on the backend).

Object instances typically refer to a group of resources that may appear two or more times on the device. For example, there could be more than one WLAN Connectivity OMA object instance. Each

instance defines a WiFi access point that the device can use to achieve global connectivity. There could also be multiple sensors of the same type on a device at the same time, such as the internal temperature sensor for health monitoring and an external peripheral sensor for environmental measurements.

5.2.2. What is Homard?

Homard is the Smart Spot **device management platform**. It is based on the OMA LwM2M protocol. The platform offers several functionalities such as remote maintenance, firmware upgrade and open/standard APIs for information reporting.

Smart Spot implements a LwM2M Client that is always connected to a LwM2M Server host in the Libelium Cloud. And Homard is the graphical tool which permits the user to manage the LwM2M Server and thus interact with the device.



Figure: LwM2M clientserver for device management

Connection to third party platforms is possible thanks to the configuration of other protocols such as MQTT, Sentilo, etc. All can be configured by the user from the Homard device management platform.

5.2.3. Access to Homard

You can access to [Homard Platform](#). Then you can login with your **User** and **Password** that is provided by Libelium.

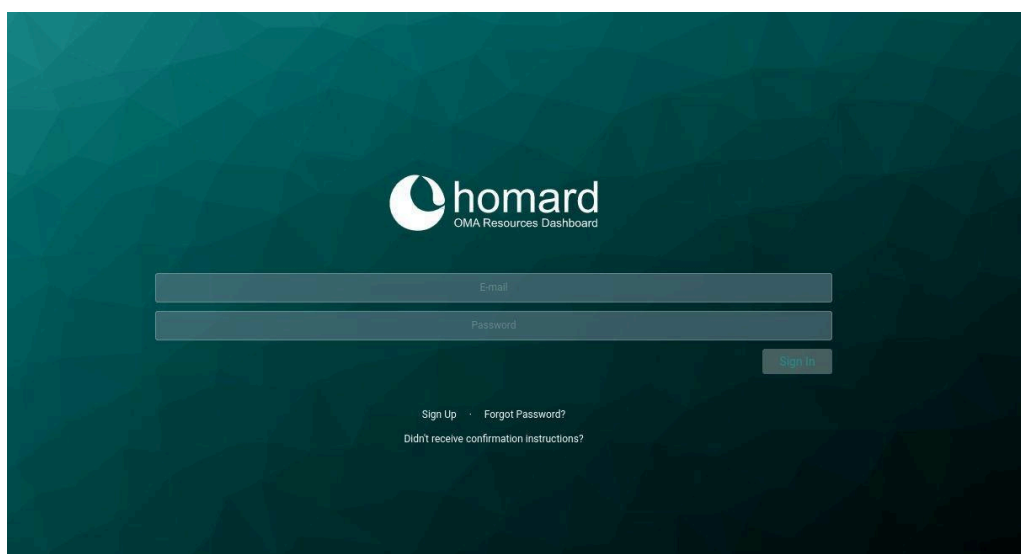


Figure: Homard login

5.2.4. Access to the device profile

It is possible to browse your different Smart Spot devices and access to each profile by clicking on the corresponding device identification:

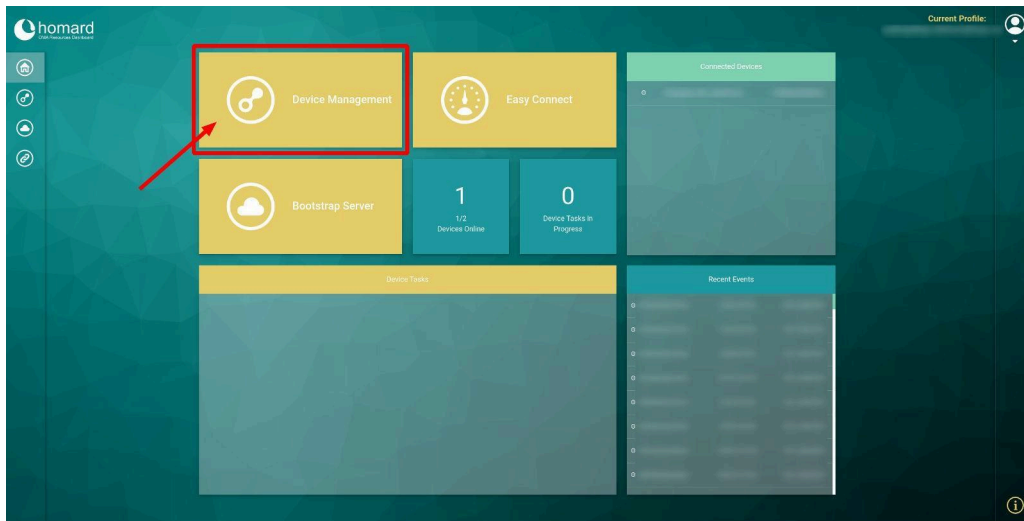


Figure: Access to Device Management

You will then see the device profile:

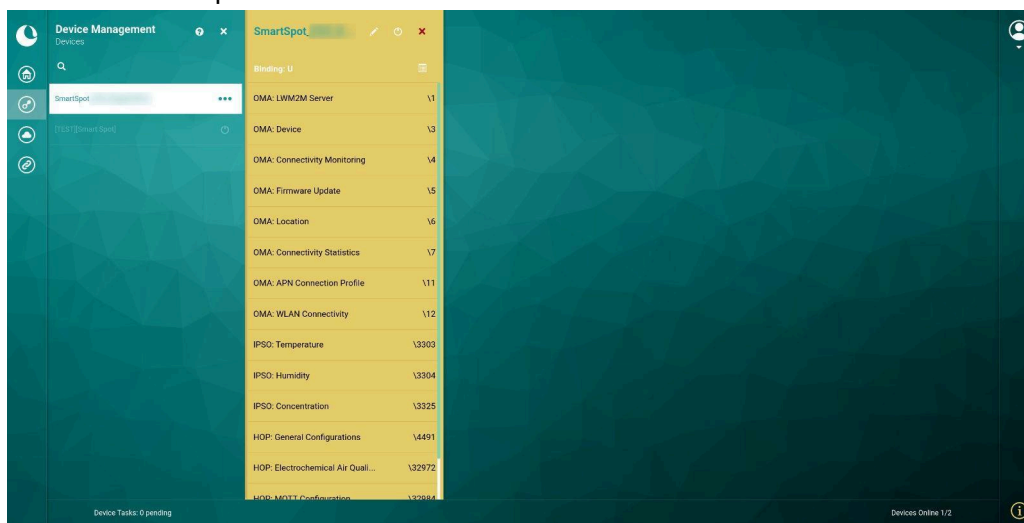


Figure: Device profile in Homard

5.2.5. Configure WiFi/Cellular connectivity (Easy Connect)

In addition to the default WiFi settings (which should only be used for the initial setup), there are two other connectivity options that should be configured in the device for a permanent connection:

- WiFi Connectivity
- APN Connectivity (SIM Card for 2G, 4G or NB-IoT)

Note 1: If you have one or more WiFi networks enabled and they are turned on, you will connect to one of them rather than via the SIM.

⚠ **Note 2:** In terms of security, once you have set up your known access method, it is possible to disable the default access point

⚠ **Note 3:** The device supports connectivity through M2M without the specifically define an APN

Homard's **Easy Connect** section allows the user to easily configure all these settings. Please access the Easy Connect section and select the connectivity you wish to configure:

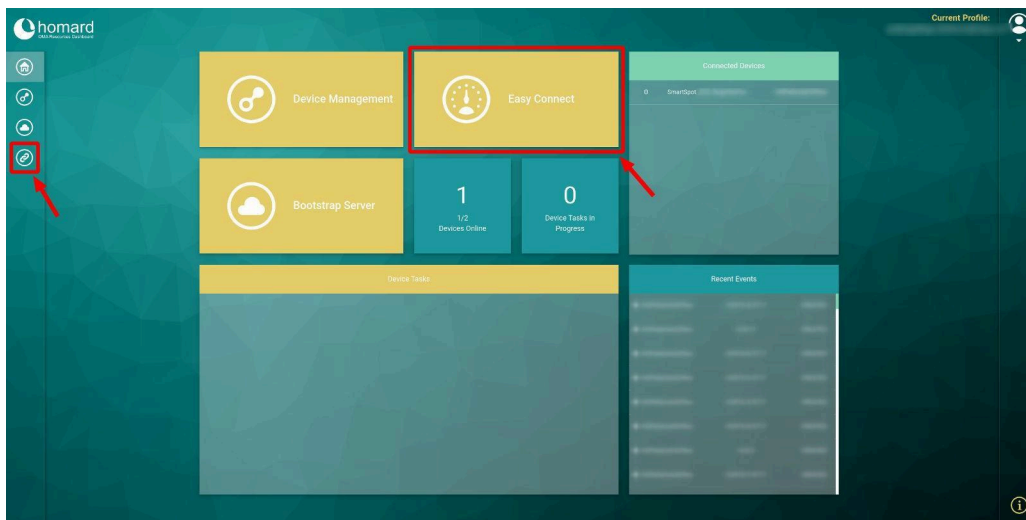


Figure: Access to "Easy Connect"

How do you want to connect the device?

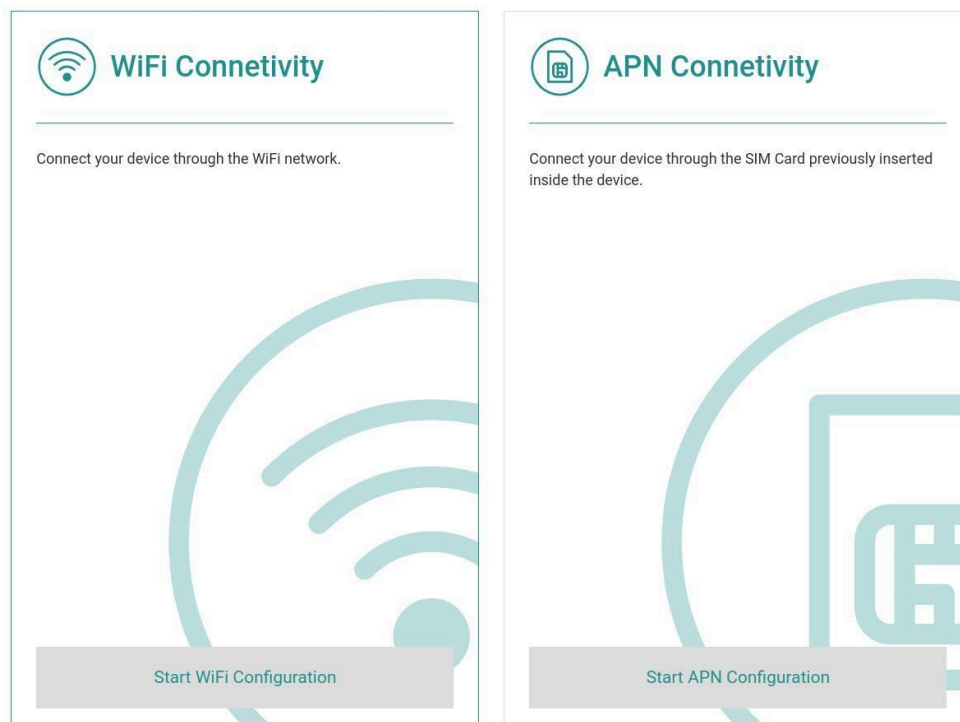
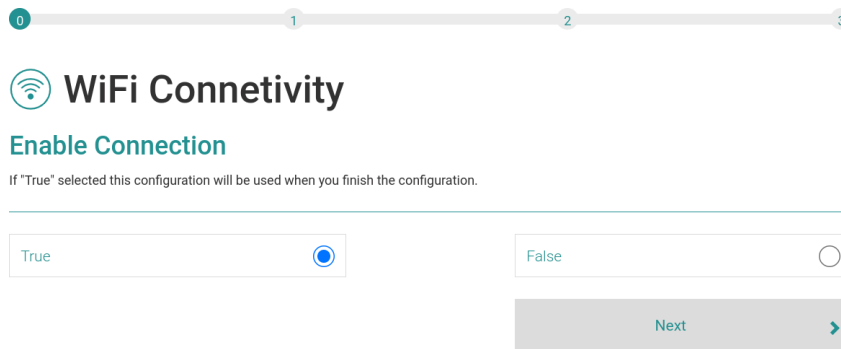


Figure: Select Connectivity option

5.2.5.1. Add a WiFi network

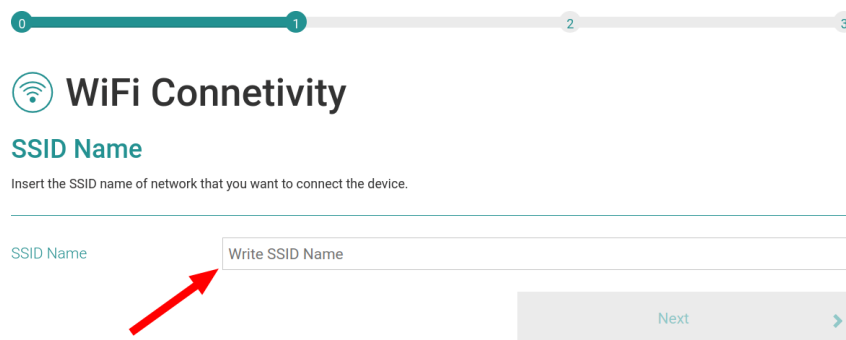
- **Step 1:** Enable WiFi connection



The screenshot shows the 'WiFi Connetivity' screen with the 'Enable Connection' section. A progress bar at the top indicates four steps, with the first step (0) being the current one. Below the title, there is a note: 'If "True" selected this configuration will be used when you finish the configuration.' There are two radio button options: 'True' (selected) and 'False'. A 'Next' button with a right arrow is at the bottom right.

Figure: Enable WiFi connection

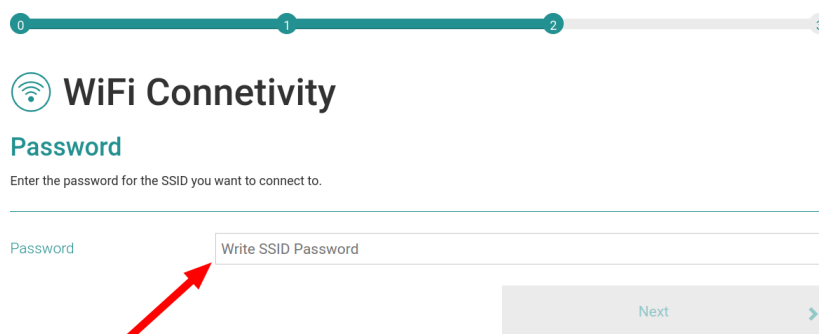
- **Step 2:** Set WiFi ESSID name



The screenshot shows the 'WiFi Connetivity' screen with the 'SSID Name' section. A progress bar at the top indicates four steps, with the second step (1) being the current one. Below the title, there is a note: 'Insert the SSID name of network that you want to connect the device.' There is a text input field labeled 'Write SSID Name' with a red arrow pointing to it. A 'Next' button with a right arrow is at the bottom right.

Figure: Set WiFi ESSID name

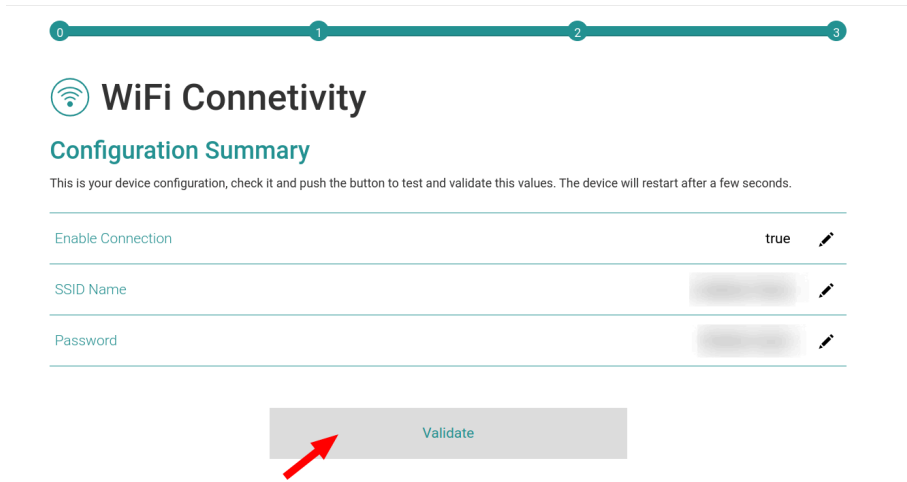
- **Step 3:** Set WiFi Password



The screenshot shows the 'WiFi Connetivity' screen with the 'Password' section. A progress bar at the top indicates four steps, with the third step (2) being the current one. Below the title, there is a note: 'Enter the password for the SSID you want to connect to.' There is a text input field labeled 'Write SSID Password' with a red arrow pointing to it. A 'Next' button with a right arrow is at the bottom right.

Figure: Set WiFi Password

- **Step 4:** Validate the new WiFi configuration



The screenshot shows the 'WiFi Connectivity' configuration summary screen. At the top, a progress bar has four steps, with step 3 being the active one. Below the title, a 'Configuration Summary' section contains a note: 'This is your device configuration, check it and push the button to test and validate this values. The device will restart after a few seconds.' The configuration details are as follows:

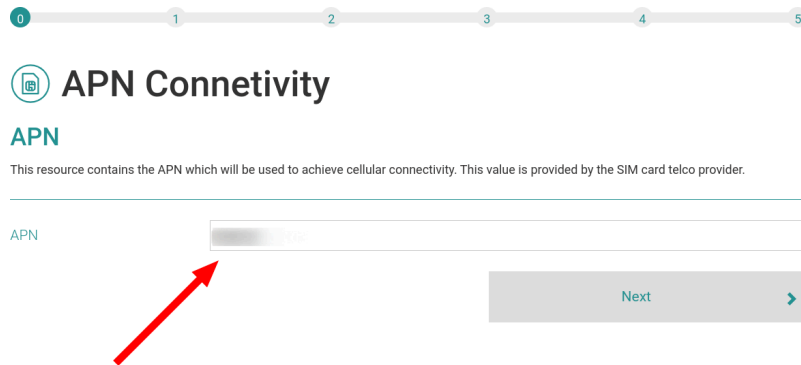
Field	Value	Action
Enable Connection	true	
SSID Name	[blurred]	
Password	[blurred]	

At the bottom, there is a 'Validate' button, which is highlighted by a red arrow.

Figure: Validate the new WiFi configuration

5.2.5.2. Configure SIM card

- **Step 1:** Set the SIM card APN



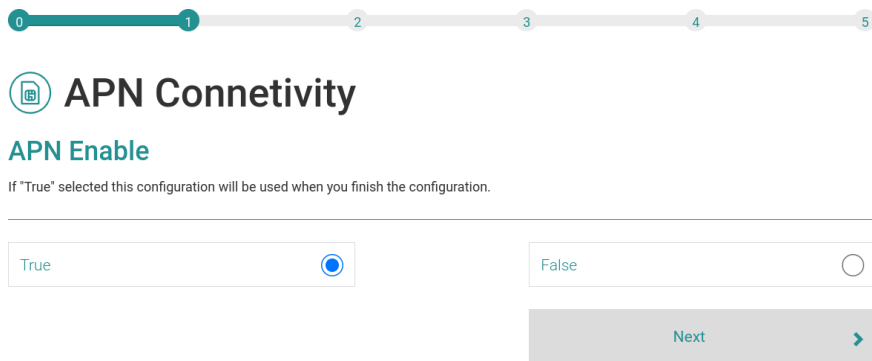
The screenshot shows the 'APN Connectivity' screen. At the top, a progress bar has six steps, with step 1 being the active one. Below the title, an 'APN' section contains a note: 'This resource contains the APN which will be used to achieve cellular connectivity. This value is provided by the SIM card telco provider.' The configuration details are as follows:

Field	Value	Action
APN	[blurred]	

At the bottom right, there is a 'Next' button with a right arrow, which is highlighted by a red arrow.

Figure: Set the SIM card APN

- **Step 2:** Enable Cellular connection



The screenshot shows the 'APN Connectivity' screen, specifically the 'APN Enable' section. At the top, a progress bar has six steps, with step 2 being the active one. Below the title, a note states: 'If "True" selected this configuration will be used when you finish the configuration.' The configuration details are as follows:

Field	Value	Action
APN Enable	True (selected)	
APN Enable	False	

At the bottom right, there is a 'Next' button with a right arrow.

Figure: Enable Cellular connection

- **Step 3:** Set APN User



APN Connetivity

User

Resource is used to specify the user name used to acces the APN. Unless specified by the telco provider, this resource could be empty.

User

Next >

Figure: Set APN User

- **Step 4:** Set APN Password



APN Connetivity

Secret

Resource is used to specify the user name used to acces the APN. Unless specified by the telco provider, this resource could be empty.

Secret

Next >

Figure: Set APN Password

- **Step 5:** Set SIM card PIN



APN Connetivity

PIN

Resource is used to introduce the PIN used to unblock the SIM card.

PIN

Next >

Figure: Set SIM card PIN

- **Step 6:** Validate the new Cellular configuration



APN Connetivity

Configuration Summary

This is your device configuration, check it and push the button to test and validate this values.

APN		
APN Enable	true	
User		
Secret		
PIN		

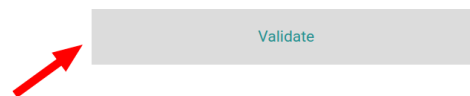


Figure: Validate the new Cellular configuration

5.2.6. Main device configurations

Through the usage of Homard device management platform it's possible to configure several parameters of the Smart Spot and carry out actions and checking. The following tables illustrate the most common resources used for device management.

Object: **OMA: Device**

Resource	Description
Reboot (/3/0/4)	Reboot the device (commonly used after apply configurations)

Object: **OMA: Connectivity Monitoring**

Resource	Description
Network Bearer (/4/0/0)	Specifies the current connectivity of the device (0=2G, 6=4G, 7=NB-IoT, 21=WiFi)

Object: **OMA: Location**

Resource	Description
Latitude (/6/0/0)	Latitude position of the device retrieved from GPS or specified on 'LBL: Device Configurations'
Longitude (/6/0/1)	Longitude position of the device retrieved from GPS or specified on 'LBL: Device Configurations'
Altitude (/6/0/2)	Altitude position of the device retrieved from GPS or specified on 'LBL: Device Configurations'

Object: **LBL: Device Configurations**

Resource	Description
General: Device mode (/32969/0/0)	Configures the device behaviour (0=M2M 1=M2M+LoRaWAN 2=LoRaWAN Sleeper,3=M2M Sleeper)
General: Interval (/32969/0/1)	Configures the measurement and reporting interval via M2M protocols. Allows setting a value from 2 to 25 minutes for device modes 0 and 1 (LoRaWAN reporting is limited by duty cycle restrictions), and from 10 to 25 minutes for device modes 2 and 3.
Location: Use fixed location (/32969/0/2)	Enables the use of a fixed latitude/longitude/altitude for the device to use instead of GPS data
Location: Specific latitude (/32969/0/3)	Configures the specific latitude
Location: Specific longitude (/32969/0/4)	Configures the specific longitude
Location: Specific altitude (/32969/0/5)	Configures the specific altitude
Web Server: Enable (/32969/0/6)	Enables the internal web server to be activated so that the user can download data in the event of areas without Internet access
[User] WiFi Access Point: Channel (/32969/0/7)	Allows to configure a channel for the WiFi access point (only necessary if the internal web server is activated).

Other information such as device information or sensor data can be found through the Lwm2m objects present on the device in Homard.

5.2.7. Device connectivity behaviour

Depending on the configuration mode selected in the 'General: Device Mode' resource of the 'LBL: Device Configurations' object in Homard, the device adapts its connectivity behaviour as described below.

Mode	Description
0 (M2M)	To allow M2M data savings (when M2M connectivity is intended to be a backup connectivity method) and due to the need to have a known source available for configuration purposes (default WiFi access point for configuration), the Smart Spot device firmware will first attempt to connect through the WiFi interface. If this is not possible because the WiFi access point does not have Internet or is unavailable, the device will try to connect through the M2M interface (2G/NB-IoT/4G). Once connected, it stays connected and reports continuously.
1 (M2M+LoRaWAN)	The device will follow the same procedure as described in the previous point, but will also attempt to connect and transmit data over LoRaWAN. This is the preferred mode for validating the LoRaWAN configuration when working with the LoRaWAN sleep mode.
2 (LoRaWAN Sleeper)	Each time the device is switched on or rebooted, it will perform the same procedure as described in the first point, so that it can be configured by the Homard if necessary. After 2 minutes, the device will automatically start to alternate between sleep and wake cycles to send data over the configured LoRaWAN network. The colour of the LED is dimmed to indicate that the device is asleep.
3 (M2M Sleeper)	The device will follow the same procedure as described in 'M2M Mode' to achieve network connectivity, but instead of staying connected and reporting continuously, it will automatically start to alternate between sleep and wake cycles to send data via the

	different M2M protocols (LwM2M/MQTT/...). The colour of the LED is dimmed to indicate that the device is asleep.
--	--

The following table illustrates the combinations of mode configuration depending on the Smart Spot model

Model	0 (M2M)	1 (M2M+LoRaWAN)	2 (LoRaWAN Sleeper)	3 (M2M Sleeper)
Crowd	Supported	Not applicable	Not applicable	Not applicable
Air	Supported	Not applicable	Not applicable	Supported
Air LoRaWAN	Supported	Supported	Supported	Supported

5.2.7.1. Behaviour mode measuring dependence

While the connected modes 'M2M Mode' (0) and 'M2M+LoRaWAN Mode' (1) are the most powerful behaviour configuration of the device, providing real-time response for management and highest measurement frequency in the sensors to provide the most accurate data, the sleeper modes 'LoRaWAN Sleeper' (2) and 'M2M Sleeper' (3) are the most energy conserving behaviour configuration of the device, where both CPU and modem are put in sleep mode for long intervals (depending on the reporting interval), reducing their consumption to the minimum order.

During sleep mode, the device keeps some sensors powered in order to maintain their readings or simply to maintain their functionality. More specifically:

- Electrochemical gas sensors (SPT-AQ-GAS-xxxx) are kept powered during sleep mode to avoid losing their stability state, as these sensors require between 4-24 hours to reach their perfect stability state after power-up. When the system wakes up, the pump introduces fresh air into the air quality sensors circuit and waits for the minimum t90 interval to elapse to ensure a correct measurement from the electrochemical sensors.
- Sonometer sensor (SPT-AQ-SEN-NLS2) remains on during sleep mode to continue noise monitoring.
- The Weather Station (SPT-SEN-DVP*2) remains on during sleep mode to continue monitoring the environmental parameters.
- The temperature and humidity sensor (SPT-SEN-TH) remains active during sleep mode as it doesn't have a significant impact on power consumption.
- The pressure sensor (SPT-SEN-PRES) is left on during sleep mode as it has little impact on fuel consumption.

The other sensors or extensions (SPT-AQ-SEN-CO2, SPT-AQ-SEN-VOC, SPT-AQ-SEN-PM, ...) are switched off in order to reduce the consumption of the system due to their consumption and their fast response when switched on.

5.2.8. Reporting interval

Based on the previous explanations of models, sensor extensions and device behaviour, the following table illustrates the minimum and maximum reporting interval that can be configured on the Smart Spot depending on these factors.

⚠ **Note:** It's important to distinguish between two types of reporting in the Smart Spot 'Crowd' model. Reporting related to general health and sensors is configured via the 'General: Interval (/4491/0/65)' resource in Homard as usual, while the reporting interval associated with detections depends on the specific configuration for 'Crowd'. See **AN20: Crowd monitoring technology and configuration via Homard** for information on configuring crowd reporting.

Smart Spot model	Hardware configuration	Mode	Minimum	Maximum
Crowd (Health + Sensors)	Any	0 (M2M)	60s (1m)	1500s (25m)
Crowd (Crowd reporting)	Any	0 (M2M)	60s (1m)	600s (10m)
Air	Without Gas or PM	0 (M2M)	60s (1m)	1500s (25m)
Air	With Gas or PM	0 (M2M)	120s (2m)	840s (14m)
Air	Any	3 (M2M Sleeper)	600s (10m)	840s (14m)
Air LoRaWAN	Without Gas or PM	0 (M2M)	300s (5m)	1500s (25m)
Air LoRaWAN	Without Gas or PM	1 (M2M+LoRaWAN)	300s (5m)	1500s (25m)
Air LoRaWAN	With Gas or PM	0 (M2M)	600s (10m)	840s (14m)
Air LoRaWAN	With Gas or PM	1 (M2M+LoRaWAN)	600s (10m)	840s (14m)
Air LoRaWAN	Any	2 (LoRaWAN Sleeper)	600s (10m)	840s (14m)
Air LoRaWAN	Any	3 (M2M Sleeper)	600s (10m)	840s (14m)

5.2.9. Device connectivity requirements

As an IoT device, the Smart Spot is able to connect to information systems or services via an Internet connection for a defined or undefined period of time. This Internet connection can be achieved through various communication protocols such as WiFi, GPRS, NB-IoT, ... depending on the use case or the customer's requirements. Due to the architecture or configuration of a network, the presence of gateways or firewalls may in some cases prevent the device from connecting.

In the case of WiFi connectivity, please note that the Smart Spot is not able to manage application layer identification from captive portals or similar, so keep its connection (MAC/IP) out of these applications.

The document **Application Note 10: Connection Requirements** specifies the connections that the Smart Spot may require to function (depending on its hardware configuration). To ensure the correct functioning of the device, check your network configuration and keep the specified ports available for the device.

5.2.10. Device geolocation

In addition to GPS availability, it is common practice to set the Latitude and Longitude parameters to specific values to avoid problems related to GNSS accuracy, GNSS induced offsets or lack of GNSS coverage, as these values would be used to define the position of the device on a map. The document **Application Note 121: Geolocation parameters configuration** will help you to perform this procedure using the Homard remote device management platform.

5.2.11. Device time synchronization

The internal clock of the device can be synchronized in three ways:

- **From the Home section of the internal web server:** This method is used when the device does not have Internet connectivity or GPS. The user can manually set the time through the WebServer interface. This must be done manually.
- **Using the GPS time:** When the device lacks Internet connectivity but has GPS, it can synchronize its clock using the time received from the GPS signal. This method provides a precision of $\pm 500\text{ms}$. This is done automatically.
- **Using SNTP protocol:** When the device has Internet connectivity, it can synchronize its clock using the Simple Network Time Protocol (SNTP). This method offers a precision of less than 250ms. This is done automatically.

5.2.12. Data Logger functionality

5.2.12.1. Introduction

In the realm of the Internet of Things (IoT), data logger functionality is critical even for devices that are typically connected to the Internet. There are several scenarios where this feature becomes indispensable:

1. **Remote Locations:** In many natural environments, such as forests, mountains or oceans, internet connectivity can be unreliable or unavailable. In these cases, a data logger ensures that all sensor data is recorded locally and can be retrieved later when the device is returned to an area with internet access, or locally via the internal web server.
2. **Intermittent Connectivity:** In environments where Internet connectivity is sporadic due to external factors such as weather conditions, network congestion or physical obstacles, a data logger ensures that no data is lost during periods of disconnection. The device can store data locally and upload it when a stable connection is re-established.
3. **Extreme Security:** In high-security environments, such as military installations or sensitive research facilities, internet connectivity may be restricted or prohibited to prevent data breaches. A data logger allows the device to operate independently of the internet, ensuring that critical data is still captured and stored securely.

By incorporating data logger functionality, IoT devices can maintain their data integrity and operational effectiveness, regardless of external connectivity constraints.

⚠ Note: This feature is always available on units with firmware v1.2 and above using device modes 0 and 3 on AQ models for the moment

5.2.12.2. Storage Robustness and capacity

For maximum robustness, speed and longevity, the Smart Spot uses a 6.5MB internal memory with binary data management, avoiding the corruption problems common in other embedded devices due to file system management.

The memory capacity depends on the number of peripherals (sensors, battery, etc.) connected to the system and the reading interval. For example, a device equipped with 6 air quality sensors, PM and VOC, with a reading interval of:

- 5 minutes, has a storage capacity of 21 days.
- 10 minutes, has a storage capacity of 43 days.
- 15 minutes, has a storage capacity of 65 days.
- 20 minutes, has a storage capacity of 86 days.

5.2.12.3. Data Retrieval methods

Data stored in the Smart Spot can be retrieved in two main ways:

- **Internet Connectivity:** Once connected to the Internet, the device transmits time-stamped stored data via embedded third-party protocols such as MQTT, HTTP or LoRaWAN.
- **Internal WebServer:** If the device is not connected to the internet, data can be retrieved by downloading a CSV file from the device's internal web server. This method ensures that all recorded data is accessible even in remote or offline scenarios. In the case of Air Quality models, it is necessary to use an additional online service provided by Libelium to process the data.

Upon downloading the CSV from the web server, make sure the file is named after the device's ID (e.g., HOPa842e388b2c2.csv). In order to obtain enhanced AQ values from the data, please open a ticket at support@libelium.com stating "Web Server AQ Data Enhancement" as the subject and attaching the CSV files containing the device data to process.

Please bear in mind this data processing service is provided ad hoc, and there may be support fees associated.

⚠ Note: To obtain the AQ enhanced data the CSV file with the AQ values must be attached to the ticket **as extracted** from the WebServer and without any modifications.

5.2.12.4. Accessing the WebServer

The device webserver is only available when the device does not have internet connectivity and it's not manageable on the Homard Device Management Platform. Otherwise, the device will directly send the data through the third party integration protocols automatically once connected.

To access the internal web server, follow these steps:

1. If you are using a smartphone instead of a computer, disable M2M connectivity from the device.
2. Search for the access point of the Smart Spot.
 - The access point name is automatically configured with the prefix 'SSAP_' and the last two digits of the device ID. For instance, a device with ID 'HOPa842e388b2c2' will generate the SSID 'SSAP_b2c2'
3. Connect to the access point and introduce the password.
 - The default password is automatically generated based on the 'ID', taking the 6 pair of digits and composing these on the following pattern:
 - L <[5]> B <[1]> L <[3]> @ <[2]> C <[4]> O <[0]> M

- For example, using the previously defined device ID 'HOPa842e388b2c2'. Where the position of its number pairs is [0]a8 [1]42 [2]e3 [3]88 [4]b2 [5]c2, the formed password is 'Lc2B42L88@e3Cb2Oa8M'
- 4. Once connected, open the browser.
 - A message indicating that the access point does not have internet, it's common and it's not a problem, as the device does not have internet connectivity.
- 5. Introduce the following url 'smartspot.local'
- 6. The webserver interface will be shown

5.2.12.4.1. WebServer managing

libelium Secure the storage. Secure the storage.		
Home	Configuration	Storage
System information Device SmartSpot AQ Model SPT-AQ-2G (BAT-PM-VOC) Framework version v5.1.3-dirty Firmware version v1.1.2-465-gb6b364cfb Device Date 1/1/1970, 1:03:40 Sync hour	Behaviour Report Interval 300 modeBehaviour Mode 1 (WiFi/M2M Connected) Apply changes	Storage information Max. registers 6312 Max. hours 526 Max. days 21 Storage usage Registers 200 Hours 16 Days 0 Storage management Download Delete

Figure: WebServer sections

As can be seen in the previous figure, the web server is divided into the following three different sections:

- **Home:** Contains information about the Smart Spot such as model, firmware version and current date.

- Provides a button to set a device date for when there is no internet connection and no GPS available.
- **Configuration:** Allows the configuration of basic settings such as measurement/reporting intervals and device mode.
 - Provides a button to set the configuration, the device will restart automatically if any of the parameters are changed.
- **Storage:** Contains information about both, the maximum storage that can be achieved with the current configuration (based on measurement/reporting interval and the size of each stored register), and the current storage in use.
 - Note that once the memory is full, the number of registers will not increase because the older registers will be discarded in favour of new ones.
 - Provides a button to download the data stored in the device.
 - Downloading a full memory can take around 30 minutes. This is not a fixed value as it depends on several variables such as the number of peripherals, the quality of the connection with the smartphone, the smartphone model, the smartphone operating system, ...
 - Provides a button to delete the data stored on the device.

⚠ **Note:** The Smart Spot exposes the access point in the 2.4Ghz band.

6. Protocols for Thirdparty data integration

6.1. MQTT

MQTT (Message Queue Telemetry Transport) is a lightweight and efficient communication protocol used for sending messages between devices connected to the Internet of Things (IoT). MQTT is based on the publish-subscribe model, where devices that generate data (publishers) send information to a central server (broker), which in turn distributes this data to other interested devices (subscribers) that have registered to receive this information. The Smart Spot uses the most widely used and stable version of MQTT (v3.1.1) and supports the two main security mechanisms to ensure secure communication between devices. These include:

- **User and password-based authentication:** MQTT devices can be authenticated using user and password credentials stored on the MQTT server.
- **Digital certificates:** MQTT devices can be authenticated using digital certificates, providing greater security and mutual authentication between devices.

Data reporting via the MQTT protocol in the Smart Spot device has always followed the philosophy of being “FIWARE Ready”, so that the data can be easily integrated into the widely used FIWARE platform, while also being compatible with any other platform not based on FIWARE.

⚠ **Note:** Due to the specific topic and message organisation of information in MQTT, the MQTT client is not directly compatible with AWS or Azure platforms.



Figure: MQTT architecture

If you are interested in integrating data via this protocol, more information about data models and device configuration via Homard is available in the specific **Application Note 13: Data Integration via MQTT** document.

6.2. Sentilo

Sentilo is an open source platform for real-time data management and monitoring in smart cities. It provides a technological infrastructure for collecting, processing and analysing data from different sources, such as sensors, IoT (Internet of Things) devices and geographic information systems.

Sentilo provides an Application Programming Interface (API) that allows developers to create customised applications and services that utilise the data collected by the platform. It also includes visualisation tools that facilitate graphical display and real-time data analysis.

The Smart Spot device has an HTTP connector that can be configured to push data to a specified Sentilo platform according to pre-defined data models.



Figure: Sentilo connector architecture

If you are interested in integrating data via this protocol, more information about data models and device configuration via Homard is available in the specific document **Application Note 16: Data Integration via SENTILO**.

6.3. LoRaWAN

LoRaWAN is a low-power wide-area network (LPWAN) protocol that is optimised for use with mobile or stationary battery-powered end-devices. A key technology in today's Internet of Things (IoT) landscape, LoRaWAN is maintained by the non-profit and open membership LoRa Alliance.

Since the first LoRaWAN specification was released in 2015, this technology has seen significant global deployment and increasing adoption by traditional cellular network operators. Designed from the ground up for IoT applications, LoRaWAN disrupts traditional wireless M2M communications by significantly reducing power consumption and operational/hardware costs (both at the end device and base station).

LoRaWAN networks adopt a star-of-stars topology made up of five different logical entity types: End-Devices and Gateways (also known as concentrators, or base stations) at the hardware level; 'Network Server', 'Application Server' and 'Join Servers' at the software level.

The LoRaWAN security architecture considers two domains: the network domain, and the application domain. It relies on the AES-128 cipher suite for data protection and packet counters for replay detection. LoRaWAN provides dynamic or static establishment of network and application session keys for authentication and encryption of network management and data communications, respectively.

Smart Spot LoRaWAN packet encoding is based on the **Cayenne Low Power Payload (LPP)**, which provides a convenient and easy way to send data over LPWAN networks such as LoRaWAN. The

Cayenne LPP complies with the payload size limitation and allows the end node to send multiple sensor data at the same time (unique transmission).

For a complete description of all messages that can be transmitted by the Smart Spot depending on the hardware configuration and services enabled on the device, please refer to the ANNEX I section of the **Application Note 19: Data Integration via LoRaWAN** document. One or more of these messages may be included in the same payload and transmitted by the device.

It's possible to configure the Smart Spot 'Air LoRaWAN' model as an End-Device to push data through both Libelium and third party gateways to Libelium or third party LoRaWAN backends for data integration.

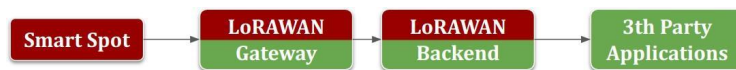


Figure: LoRaWAN architecture

⚠ **Note:** This communication method is ONLY available for the 'Air LoRaWAN' model

7. Software services for Smart Spot

In addition to the device, Libelium also offers specific platform deployment services in order to exploit data that can be used in conjunction with the Smart Spot device or independently with other Libelium devices.

The deployed platform provides two ways to interact based on the client's needs:

- Smart city viewer based on Grafana. It provides a graphical interface for both technical and non-technical users.
- Programming interfaces. The platform provides a range of interfaces for device integration, data consumption and authentication management.
 - Device provisioning API
 - Context data API
 - Historical data API

All services have the following key features

- Minimal resource exposure.
- Communicate over a private network for microservices.
- All external communications are secured by digital certificates (TLS).
- Secure connection of IoT devices using protocol-specific mechanisms. MQTTs with certificates.
- Data isolation within the platform.
- Ability to create isolated data spaces for different clients or verticals within the platform.
- Centralized authentication and authorization through an identity provider.
- Widely used security protocols such as OAuth2.0/OpenID Connect.
 - Single Sign-On
 - Integration with major identity providers (Google, Github) and directory solutions (LDAP, ActiveDirectory)
 - Customization of the login flow (2FA)
 - Role-based, attribute-based, or mixed security.
- Auditing through service logging. Deployment of ELK stack.

The following image illustrates the different components involved in the platform.

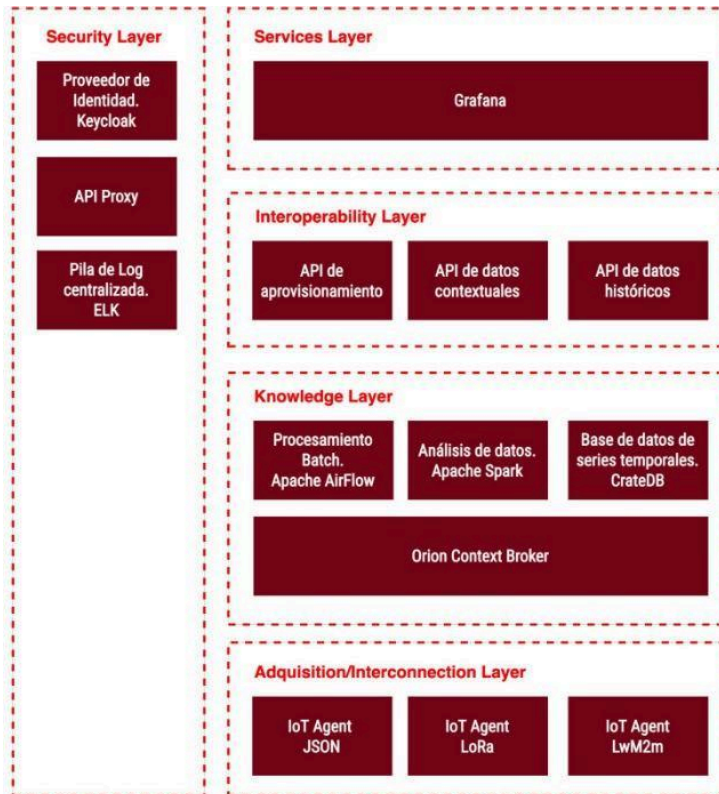


Figure: Server services architecture

This platform is deployed on demand by the Libelium team and can be located in different locations depending on the customer's needs.

7.1. Libelium FIWARE Saas

The platform deployment is centralized as a service in the Libelium Cloud using Docker containers and Ansible. This approach allows for agile updates and the addition of new functionality to the platform.

7.2. Libelium FIWARE On Premise

On-demand deployment in the customer's data centre (CPD) using Docker containers and Ansible. Adapted to resources based on the solution required.

7.3. Libelium FIWARE Private Server

On-demand deployment in the cloud using Docker containers and Ansible on a dedicated private server.

7.4. Visualizers

Libelium offers visualisers for all FIWARE scenarios (SaaS, ON Premise or Private Server). Depending on the application deployed, the visualizers adopt a different approach (Crowd monitoring, Air Quality pollution, Noise Level monitoring, etc.).

Customised visualisers can also be developed to meet specific customer requirements. Please contact Libelium for a quotation.

7.4.1. Air Quality

Regarding Air Quality sensors, it is possible to ask Libelium for dedicated visualizers such as:

- Maps with main indicators:

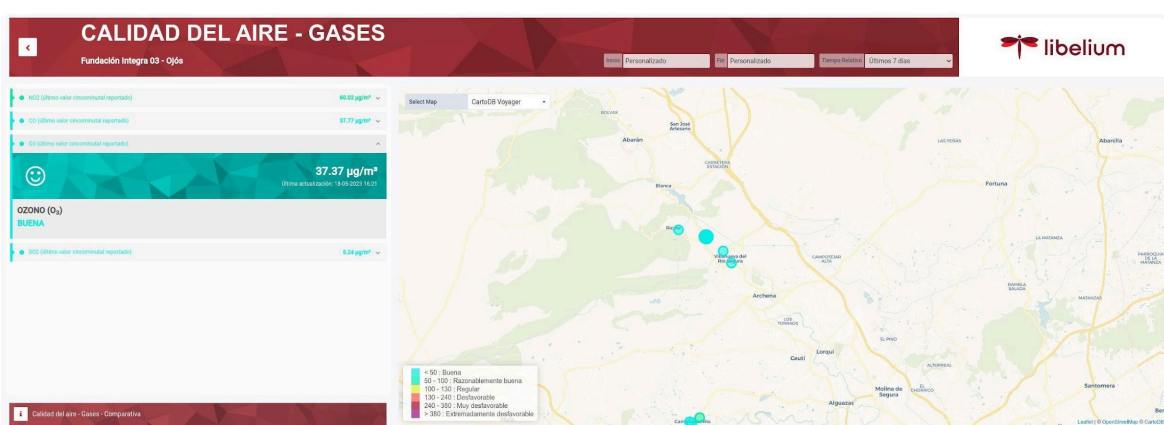


Figure: Map

- Pollutants level evolution along time:

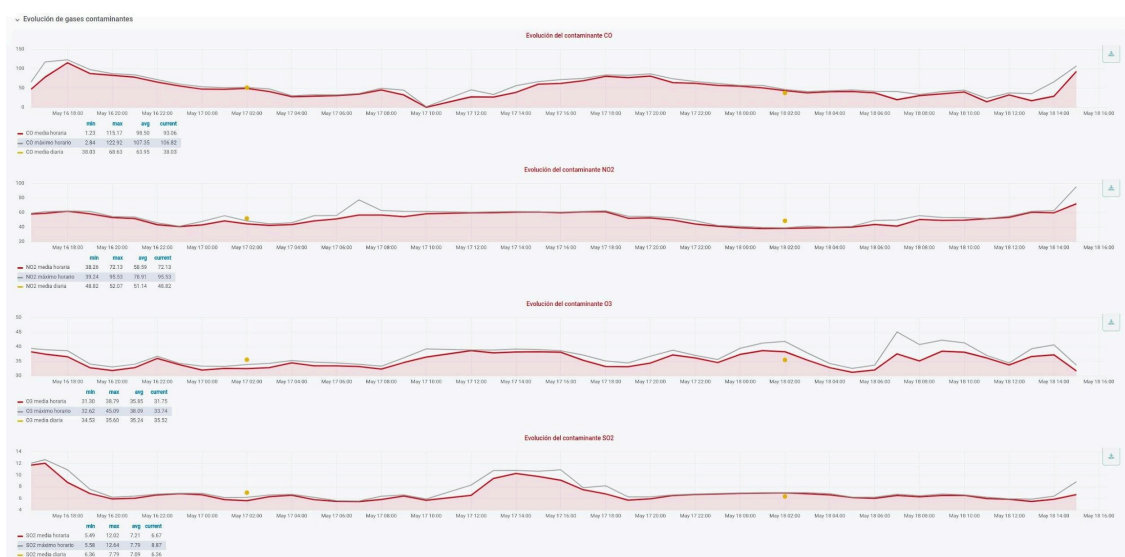


Figure: Pollutants level evolution

- AQI index represented as coloured frames within the graph:



Figure: AQI index coloured

- Particulate Matter map and indicators:

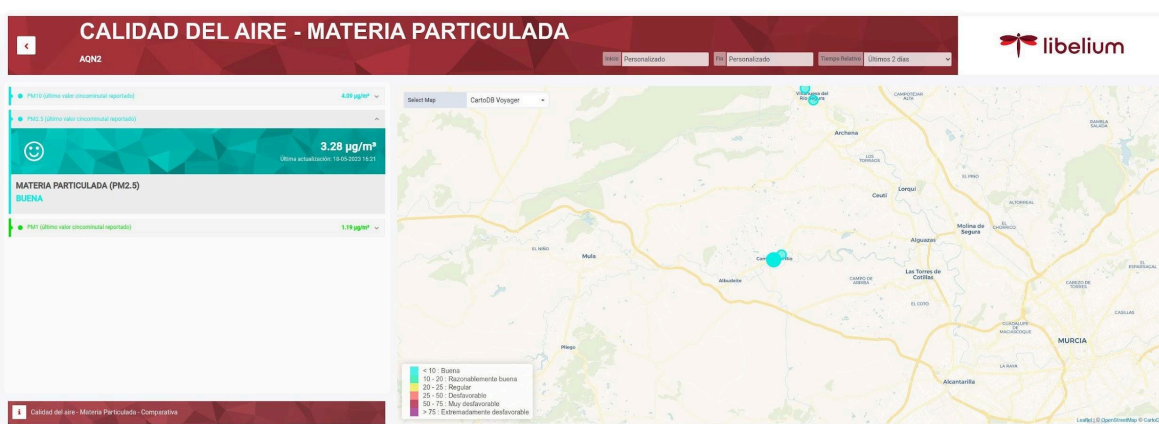


Figure: Particulate matter map

- Particulate Matter evolution (PM1, PM2.5 and PM10):

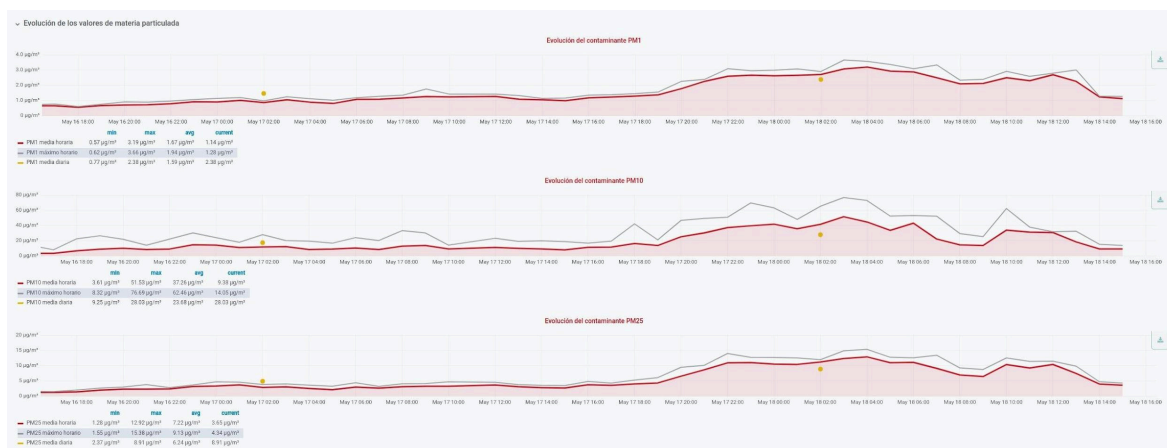


Figure: Particulate matter evolution

- Download all kind of data (CSV, JSON or TXT):



Figure: Download data

7.4.2. Crowd monitoring

Regarding Crowd monitoring services, it is possible to ask Libelium for dedicated visualizers such as:

- Count service in map with main indicators:

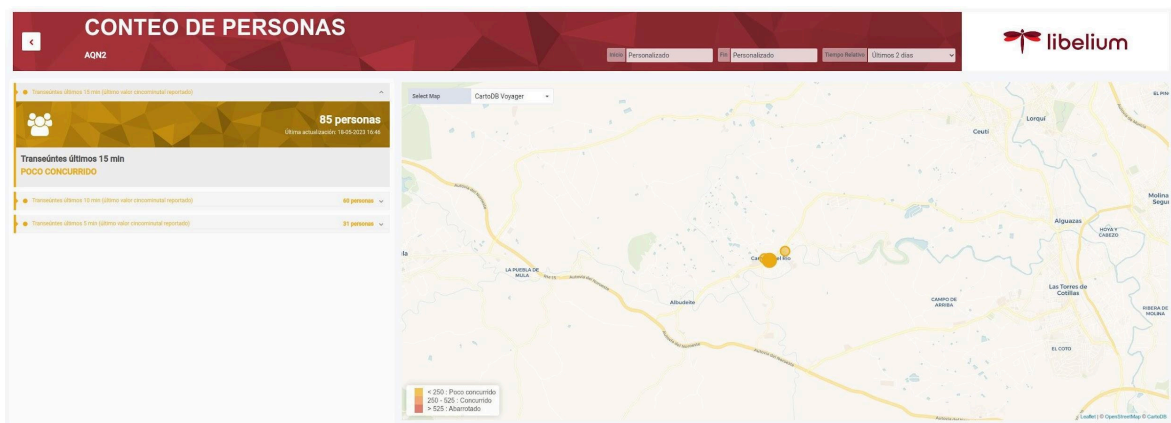


Figure: Crowd count service in map

- Count service evolution along time:



Figure: Evolution of Crowd count service

7.4.3. Noise Level

Regarding Noise Level sensors, it is possible to ask Libelium for dedicated visualizers such as:

- Maps with main indicators:

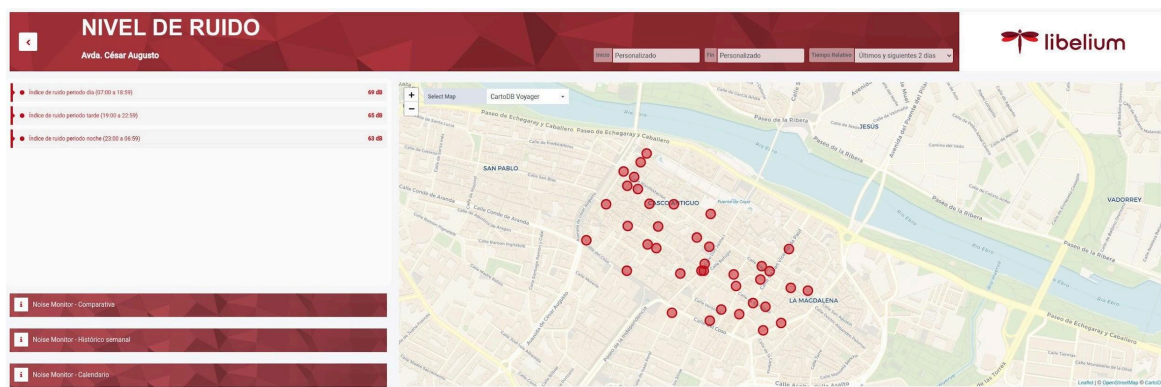


Figure: Map

- Acoustic index values (LAeq, Lden, Ld, Le, Ln, L1, L10, L50, L90, L99):



Figure: Index values

- Weekly history:

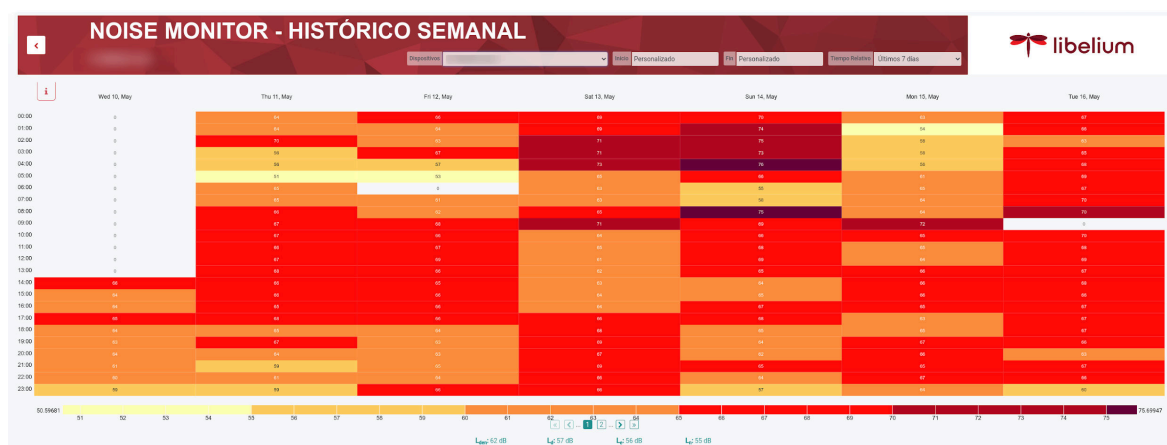


Figure: Weekly history

- Hourly aggregated data :



Figure: Hourly data

7.5. Maintenance services

Maintenance is always included (*):

- Access to the Libelium Homard device management platform
- Software/Firmware Updates (**)
- Customer service through ticketing system for incident assessment
- Data Quality Improvement Service: Automatic application of AI and ML algorithms to process, update and continuously improve device data (Data Quality Assessment)

(*): This service does not include sensor replacement or removal/reinstallation costs (these are budgeted separately).

(**): Corrective maintenance

8. Maintenance

8.1. General maintenance

General recommendations:

- Although the Smart Spot is a highly resistant product, please treat it with care to prolong its life.
 - Handle the Smart Spot node carefully, do not drop it or move it roughly.
 - Avoid placing the devices in areas where high temperatures may occur, as this may damage the electronic components.
 - Avoid aggressive environments (corrosive gases, animals, excessive/condensing humidity or atmospheres with salt suspended in the air).
 - Do not force the antennas onto their connectors as this may damage the connectors.
 - Only plug antennas or extension probes into their appropriate sockets.
 - Do not use any type of paint on the device as this may affect the operation of the measurements, connections and locking mechanisms.
 - To avoid damage to the electronic components, do not store the Smart Spot in places exposed to dirt and dust.
 - Use a damp cloth for cleaning, do not use aggressive chemical products.
 - The height of the installation should be at least 4m.
 - Ordinary personnel should not have access to the device (external nor internal).
 - Only expert operators should have access to the device for maintenance or calibration purposes.
 - This equipment is not suitable for use in locations where children are likely to be present.
- Recommended annual maintenance:
- Verify that the post and/or the device is aligned and make adjustments if needed.
 - Verify that the Outdoor electrical panel is aligned and make adjustments if needed.
 - Check the cables for any damage.
 - Check the solar panel installation and make adjustments if needed.

8.2. Weather station maintenance

Recommended annual maintenance:

- Verify that the post is aligned and make adjustments if needed.
- Ensure the rain gauge is balanced and adjust it if necessary.
- Wipe down the temperature and humidity sensor with an alcohol wipe.
- Remove debris screen and clean out the debris in the cone. For greatest accuracy, clean your rain collector a few times a year.
- Check the sensor cable for any damage.
- Ensure the wind sensor arm is correctly aligned to point north.
- Verify that the wind vane rotate freely.
- Make sure the wind vane can rotate without obstruction.

8.3. Air Quality extension

The document **AN132: Air Quality sensors maintenance** provides specific steps for keeping maintenance of all embedded sensors in the air quality extension:

- Particle Matter
- NO2

- O3
- CO
- SO2
- NO
- H2S
- NH3
- CO2
- VOC
- Noise level sensor.

Although the **Sound Level Meter** is factory calibrated, users may wish to recalibrate the Sonometer periodically for maintenance and verification purposes. The document **AN131: Sonometer maintenance procedures** provides specific steps for this procedure.

8.4. Wind anemometer and direction sensor

Recommended periodic maintenance:

- Ensure that the mounting structure for both the anemometer and wind vane is stable and secure. Check for any signs of loosening or damage that could affect the sensor's performance.
- Check the cables for any damage.
- Ensure the wind sensor indication is correctly oriented to point north.
- Ensure it is freely rotating and not obstructed by any external objects.
- Clean the anemometer cups and wind vane using a soft brush or cloth to remove dirt and dust. Avoid using abrasive materials or strong chemicals.
- Inspect and replace any worn or damaged components such as cups, bearings, wind vane elements, or any seals.

8.5. AWDTHP sensor

Recommended periodic maintenance:

- If the sensor is installed in a metal pole, verify the post is aligned. Make adjustments if needed.
- Verify the sensor is aligned. Make adjustments if needed.
- Check the cables for any damage.
- Ensure the wind sensor indication is correctly oriented to point north.
- Inspect and replace any damaged parts.

8.6. Temperature and humidity sensor

Recommended periodic maintenance:

- If the sensor is installed in a metal pole, verify the post is aligned. Make adjustments if needed.
- Verify that the sensor is correctly aligned and securely mounted.
- Ensure the solar shield is properly oriented to minimize direct sunlight on the sensor.
- Check the cables for any damage.
- Clean the solar shield and sensor using a soft brush or cloth. Avoid using cleaning solutions that could damage the sensor.
- Check for any insects, dirt, or other contaminants inside the solar shield that could affect sensor readings

- Use a portable temperature and humidity meter to compare readings. Ensure the sensor is functioning within the specified accuracy range.
- Inspect and replace any worn or damaged parts.

9. Disclaimers

9.1. Enclosure modification

The application of paint or any surface-altering substance to this device is undertaken solely at the discretion and risk of the user. Such modifications may interfere with the functionality or accessibility of essential components, including but not limited to electrical connectors and fastening screws. Any alteration of this nature may compromise the integrity of the device and will render the product warranty null and void. The manufacturer disclaims all liability for damage, malfunction, or performance issues arising from unauthorized surface treatments.

9.2. Customer-provided SIM connectivity

Important notice regarding customer-provided SIM cards with Smart Spot.

At Libelium, we ensure the reliability of our Smart Spot device, especially with our global SIM card. We have optimized Smart Spot firmware and configuration for seamless performance. While we recommend our SIM card for the best experience, we understand some customers prefer their own.

Please be aware that using your own SIM may lead to connectivity and configuration issues. All Smart Spot devices could become unresponsive, so the only solution would be to re-flash new firmware with cable.

We allow the use of customer-provided SIMs under specific conditions. By choosing to use your SIM, you accept the responsibility for any potential problems that may arise.

To minimize risks, customer SIM cards and networks must meet the following criteria:

- Static and known APN configuration.
- The APN must not require advanced or proprietary authentication methods.
- Keep in mind that firewalls may prevent communication.
- Multi-network technology support depending on the model (2G/3G/4G with fallback in Smart Spot 4G model) and NB-IoT in NB-IoT model.
- No SIM locking or carrier restrictions.
- SIM/Network does not block necessary communication ports.
- No Aggressive Traffic Filtering or Shaping.

We strongly advise carefully considering this information before using your SIM with Smart Spot. Before any field deployment, you must thoroughly study your SIM configuration and perform extended testing of your SIMs with Smart Spot at your facilities over multiple weeks.

Kindly understand that meeting these criteria helps reduce—but does not eliminate—the risk of failure.

10. Disposal and recycling

- When Smart Spot reaches the end of its useful life, it must be taken to a recycling facility for electronic equipment.
- The equipment should be disposed of separately from solid urban waste, please dispose of correctly.
- Your distributor will advise you on the most appropriate and environmentally friendly way to dispose of the product and its packaging.

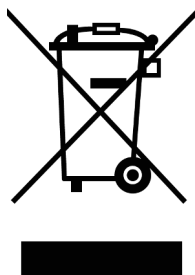


Figure: The Smart Spot must be recycled

11. Documentation changelog

v1.18: July 2025

- Added disclaimer about SIM connectivity
- Align "Device Configuration" resources to current firmware release.

v1.17: June 2025

- Added resource to activate the internal web server.
- Added resource for configuring the WiFi access point channel.

v1.16: June 2025

- Constrained reporting intervals for Air models to ensure correct gas data correction.

v1.15: May 2025

- Disclaimer added to avoid painting the device

v1.14: May 2025

- Included Noise Level sensor as option for the Smart Spot Crow Monitor model.
- Added installation instructions for the Noise Level sensor for Smart Spot Crowd Monitor model.

v1.13: Mar 2025

- Reduced the number of outdoor power supply models to only one
- Typo correction

v1.12: Jan 2025

- New format
- Typo correction

v1.11: Dec 2024

- Added Gas Sensors as optional in 'Air' and 'Air LoRaWAN' models
- Mode availability table added for different models
- Added section on minimum and maximum configurable reporting interval

v1.10: Sept 2024

- Improved instructions and images added for solar panel extension
- Improved Core installation instructions
- Improved instructions for Air Quality Extension installation
- Improved instructions regarding temperature/humidity sensor installation
- Added information about the need to use the 2.4GHz band when configuring an access point for Smart Spot

v1.9: Aug 2024

- Added information regarding usage of sensors in sleep mode
- Added information about new Solar Panel extension
- Added information about new Extreme battery
- Added maintenance recommendations

v1.8: Jul 2024

- Added RAEE logo on “Disposal and recycling” section

v1.7: Jun 2024

- Added new AWDTHP sensor

v1.6: Jun 2024

- Added a new device mode (M2M Sleeper)
- Added note to specify the ability to use M2M without configuring an APN
- Added information about LED behaviour in sleeper mode.
- Added information on reporting PM4, TSP and TPC values.

v1.5: May 2024

- New end-less clamps for multipurpose installation
- Fix errata

v1.4: April 2024

- Improvements to the definition/explanations for the ‘Crowd’, ‘Air’ and ‘Air LoRaWAN’ models
- Updated images for new core encapsulation in several sections
- Added and improved information on the installation section
- Full document review (typo and improvements)

v1.3: March 2024

- Update Size/Weight/Material to new core encapsulation
- Update pictures to new core encapsulation
- Move to ‘Crowd MAC/SSID reporting’ naming

v1.2: November 2023

- Fix errata
- Fix Modbus TCP reference

v1.1: July 2023

- Fix gas sensor cross sensitivity table

v1.0: June 2023

- First version of the document